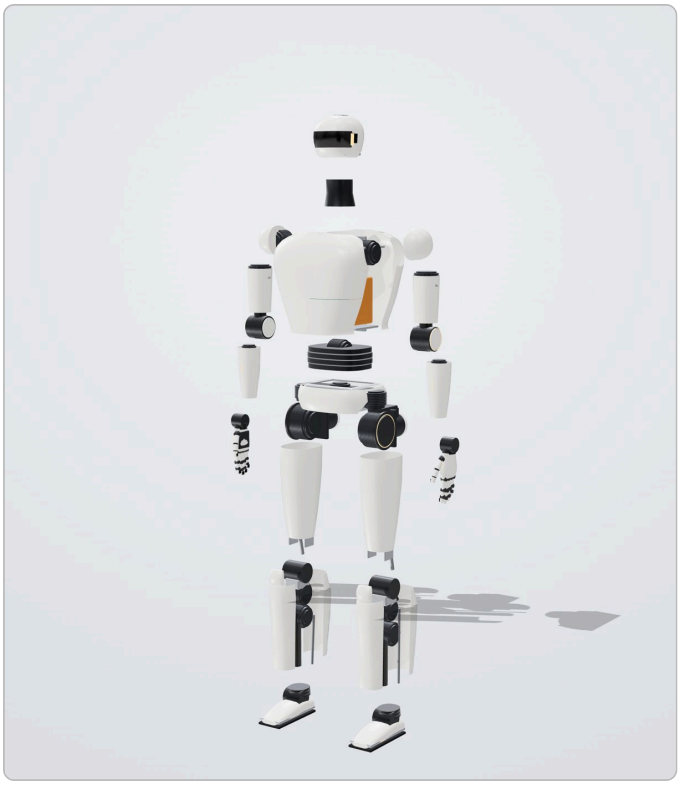
▼ PHASES AND APPENDICES

P00	Workshop, safety and documentation setup
P01	Test infrastructure: gantry, dynamometer, e-stop chain, bench power
P02	Electronics bring-up: RT computer, EtherCAT, drive carrier, safety MCU
P03	KA-90P and KA-110P planetary actuators (first articles, then series)
P04	KH strain-wave actuators (KH-25, KH-20, KH-14)
P05	KM-130 custom motors and KA-130K / KA-130H actuators
P06	Battery pack 16S6P with BMS and solid-state disconnect
P07	Legs: hip, thigh with knee four-bar, shank with ankle push-rods, foot
P08	Pelvis, waist, torso frame, power distribution and compute
P09	Arms: shoulders, upper arms, elbows, forearms, wrists
P10	Hands (10 actuated DOF each)
P11	Neck, head, cameras, LiDAR, calibration
P12	Harness, cooling, shells and final assembly
P13	Software bring-up and commissioning hand-off
App. A	Testing and commissioning gates
App. B	Manufacturing package
App. C	Optional: scratch-built motors and actuators

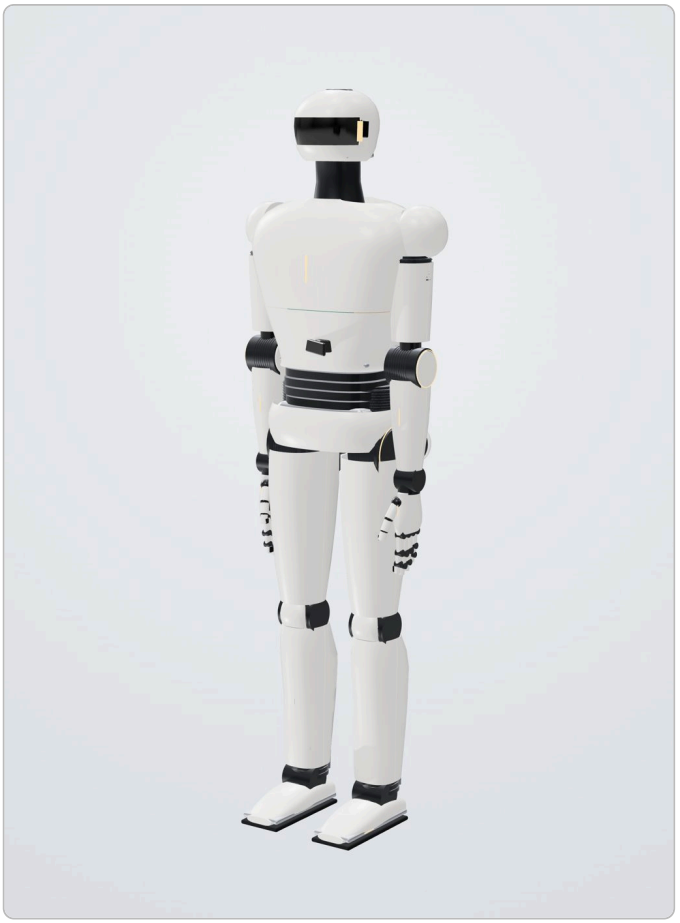
FULL-SIZE HUMANOID · 1.76 M · 84 KG ESTIMATED (92.6 KG DESIGN MASS) · 16S LI-ION

KESTREL K1 build tutorial

Every phase from an empty workshop to a commissioned robot. In each phase: visual reference, parts with BOM IDs, tools, manufacturing, assembly with torques, electrical work, and the bench test that has to pass before you move on. 15,856 words, 44 figures and drawing sheets.



RENDER Exploded assembly



RENDER Fully assembled

This is the step-by-step build guide. Work through the phases **in order**. Each phase ends in a test gate (docs/11). Do not start a phase until the previous gate has passed and been logged.

How each phase is laid out

Section	Contents
A. Visual reference	renders, cutaways, diagrams, CAD files and drawing sheets for this phase
B. Parts list	every part with its BOM ID (`data/bom.csv`), quantity, specification, make/buy
C. Tools & equipment	what you need on the bench
D. Manufacturing	how to make the custom parts: material, process, tolerances, finishing
E. Assembly	ordered steps with orientation, fasteners, torques, adhesives, alignment
F. Electrical	wiring, connectors, pinouts, voltages, precautions
G. Testing	the bench test that proves the assembly before it goes into the robot, with pass criteria

Phase map

Phase	What you build	Gate
P00	Workshop layout, lithium safety, electrical safety rules, documentation system	—
P01	Gantry + fall-arrest harness, actuator dynamometer, e-stop chain, bench power	—
P02	RT computer + EtherCAT master, drive carrier PCB, safety MCU, one drive spinning one motor	G0
P03	KA-90P (first article), then KA-110P: planetary actuators with commercial frameless motors	G1
P04	KH-25 / KH-20 / KH-14 strain-wave actuators with torque flexures	G1
P05	KM-130 custom motors and KA-130K / KA-130H actuators	G1
P06	16S6P battery pack, BMS, solid-state disconnect	G0 (pack)
P07	Hip, thigh with knee four-bar, shank with ankle push-rods, foot	G2
P08	Pelvis, waist, torso frame, power distribution, compute	G2/G3
P09	Shoulders, upper arms, elbows, forearms, wrist linear actuators	G2
P10	Hands: palm actuators, finger linkages, tactile sensors, controller	G2
P11	Neck, head frame, cameras, LiDAR, calibration	G3
P12	Harness, cooling ducts, shells, final assembly and inspection	G3
P13	Software bring-up and progressive commissioning (hand-off to docs/11 G3–G9)	G3–G9

Why this order

1. **Rigs and safety before anything moves.** The gantry, e-stop chain and dyno are the tools that make every later test safe and measurable.
2. **One of each actuator class first.** The first article proves the design (Kt, efficiency, thermal behaviour, backlash) before you buy 13 more gear sets. Results feed back into `calc/kestrel/actuators.py`.
3. **Simplest risk first.** KA-90P (commercial motor, simplest planetary) comes before the custom KM-130 motor.
4. **Limbs before the body.** Each limb is tested clamped on a stand before it carries the robot's weight.

Non-negotiable safety rules (repeat them to yourself every session)

- The pack and the service disconnect are **out** before you touch any connector, shell or harness.
- No powered motion of a limb or the robot without the **e-stop in hand** and a second person present for G3 onward.

- No walking tests without the **gantry harness** attached until gate G6 says otherwise.
- Charge batteries only when supervised, on a non-flammable surface, in the battery area.
- Spring-loaded or magnetised parts (rotors, magnets) are handled with jigs, never fingers.
- High-torque joints are pinch hazards. Keep hands out of joint envelopes whenever drives are enabled, even in "zero torque".

P00 Workshop, safety and documentation setup

A Visual reference

- Safety architecture: `diagrams/safety_chain.svg`
- Test progression: `diagrams/test_gates.svg`

Suggested floor plan (minimum ≈ 6 × 5 m):

```
+-----------------------------------------------------------+
|  BATTERY AREA (concrete floor, metal cabinet,  |  BENCH 1: electronics / soldering   |
|  sand bin, extinguisher, smoke detector)       |  (ESD mat, fume extraction)         |
|------------------------------------------------+-------------------------------------|
|  BENCH 2: mechanical assembly, surface plate   |  DYNO (bolted to bench, guard box)  |
|------------------------------------------------+-------------------------------------|
|  TEST CAGE 4 × 3 m under the 2 t gantry (mesh panels 1.2 m, foam floor tiles)        |
|  operator position outside the cage, e-stop on a lanyard                              |
+-----------------------------------------------------------+
```

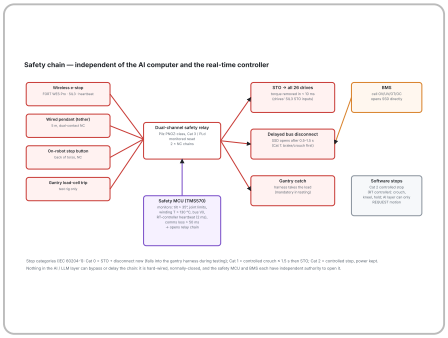


DIAGRAM Safety architecture



DIAGRAM Test progression

B Parts list

BOM	Item	Qty
—	Metal flammables cabinet for cells/packs, smoke detector, Class D or large AVD/water-mist extinguisher, fire blanket, sand bin	1 set
—	ESD mat + wrist straps, fume extractor	1 set
—	Labels, label printer, parts bins	—
B083	Gantry + harness (built in P01)	1

C Tools & equipment

See docs/12 §4 (instruments, hand tools, metrology).

D Manufacturing

None.

E Setup steps

1. Set up the **battery area** at least 2 m from exits and from flammables. Put a smoke detector over it and an extinguisher by the door.
2. Fit **ESD protection** at the electronics bench: drives, encoders and the Thor carrier are ESD-sensitive.
3. Start a **build log**: a git repository with `logs/YYYY-MM-DD.md`, every measurement, photos, part serial numbers, torque values and test results. Calibrations are versioned here (docs/07 §5).
4. Print the **gate checklist** (docs/11 §1) and sign each gate.
5. Agree the **two-person rule** with a helper for every powered test from G3 onward.

F Electrical safety rules

- The robot bus is 48–67.2 V DC: extra-low voltage, but above the 60 V touch-safe level and able to deliver kiloamps into a short. Treat every bus conductor as live until measured below 5 V.
- Before working on the robot: open the manual service disconnect, remove the pack, then wait 1 minute (bus capacitors discharge through the bleed resistors on the PDU). Verify < 5 V with a meter.
- Use insulated tools for all battery work. Remove rings and watches.
- Never short-circuit a cell or pack "to test it".

G Check

Walk through a dry run of a battery fire response and an e-stop response with your helper. Log it.

P01 Test infrastructure: gantry, dynamometer, e-stop chain, bench power

A Visual reference

- Safety chain: `diagrams/safety_chain.svg`

Dynamometer layout (top view):

```
[absorber motor + drive]──[jaw coupling]──[FUTEK TRS605 torque transducer]──[jaw coupling]──[ACTUATOR
       bolted to 25 mm Al tooling plate, both shafts aligned to 0.05 mm, polycarbonate guard box over
       reaction: actuator housing bolted to an angle plate through its base flange pattern (docs/09 §
```

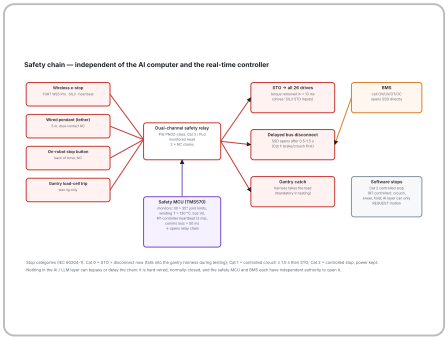


DIAGRAM Safety chain

B Parts list

BOM	Item	Qty	Spec
B083	Overhead gantry 2 t + trolley	1	≥ 2.4 m clear under the beam, beam ≥ 4 m
B083	Fall-arrest harness for the robot: chest + pelvis straps tied to lifting points on the torso frame and pelvis bowl, 6 mm Dyneema, shock-absorbing lanyard, chain hoist for height	1	rated ≥ 10× robot mass
B084	FUTEK TRS605 inline torque transducer (1–1000 N·m class, encoder)	1	via Metromatics (AU)
B084	Absorber: a spare KA-130-class motor + drive, or a hysteresis brake ≥ 450 N·m peak via a 1:1 coupling	1	regenerates into the bench supply / brake resistor
B084	Jaw couplings, 25 mm tooling plate, angle plate, guard box	1 set	
B043	FORT Wireless E-Stop Pro	1	
B044	Safety relay + wired pendant	1	
—	Bench supply ≥ 70 V / ≥ 10 A programmable + DC load + 8 Ω 100 W brake resistor on a clamp board	1 each	

BOM	Item	Qty	Spec
B085	Single-leg stand: vertical linear rail (HGR20 class) carrying a hip mount, base with 4 load cells	1	

C Tools

Drill press, tap set, hoist, laser level, dial indicators (shaft alignment), torque wrench.

D Manufacturing

- **Dyno plate:** 25 mm 6061 tooling plate with an M8 grid at 50 mm pitch. Mount the angle plate with the KA-130 (PCD 125), KA-110/90 and KH base-flange patterns.
- **Robot lifting points:** two M10 lifting eyes, one on the pelvis bowl and one on the spine plate top rib. Both are designed in the CAD; check thread engagement ≥ 1.5 d in 7075.
- **Leg stand:** carriage plate with the hip-yaw base pattern. The rail lets the hip move vertically only, and a pneumatic counterweight (or bungee) offsets 70 % of the body weight.

E Assembly

1. Install the gantry per its manual. Load-test it at 1.25× the robot design mass (115 kg) with sand bags for 10 minutes.
2. Assemble the dyno line. Align the shafts to ≤ 0.05 mm TIR and ≤ 0.1° angular, using dial indicators on the coupling hubs.
3. Wire the **e-stop chain** on a test board: FORT receiver + pendant + panel button in series → safety relay (dual channel) → two outputs: (a) STO pair to the drive under test, (b) enable input of the bench disconnect.
4. Add a **brake resistor clamp** (comparator at 70 V on / 68 V off driving a 100 V MOSFET) across the bench bus. It absorbs regeneration when the supply can't sink current.

F Electrical

- Dyno bus: bench supply → 30 A fuse → bench contactor/disconnect (enable = safety relay) → bus with clamp → drive under test + absorber drive.
- Torque transducer → its amplifier → USB/analog → a logging PC (≥ 1 kHz). Log actuator current, speed, temperature and torque together.

G Testing

Test	Pass
E-stop: press each device while a test motor spins at 50 % speed	torque removed < 10 ms (scope the STO line and the phase current); latched until a deliberate reset

Test	Pass
Wireless e-stop out of range / battery removed	stop within the configured heartbeat timeout; measure and log it (not published, R2)
Dyno torque calibration: 1 m lever arm + calibrated weights at 10, 50, 100 N·m	transducer within ±0.5 % FS
Gantry harness drop test with a 93 kg sand dummy from 0.3 m slack	arrest without the dummy touching the floor; lanyard deploys as rated

P02 Electronics bring-up: RT computer, EtherCAT, drive carrier, safety MCU

Gate G0

A Visual reference

- Comms: `diagrams/comms_topology.svg`
- Power: `diagrams/power_architecture.svg`
- Software layers: `diagrams/software_stack.svg`

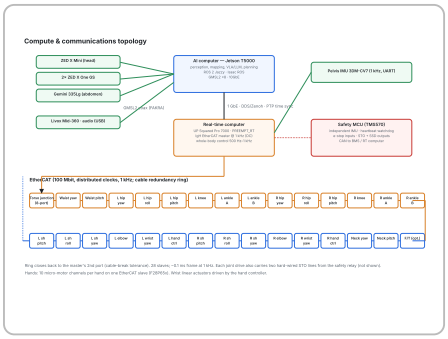


DIAGRAM Comms

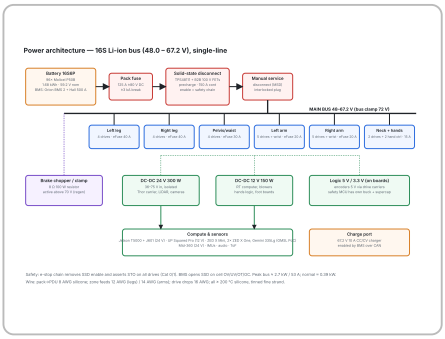


DIAGRAM Power

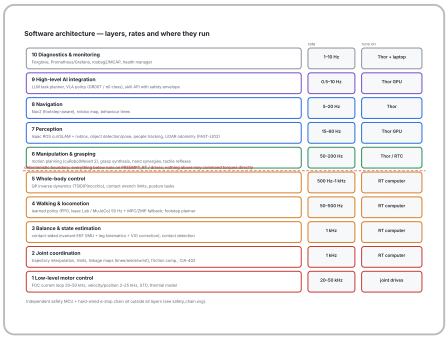


DIAGRAM Software layers

B Parts list

BOM	Item	Qty (now)
B041	UP Squared Pro 7000 (i3-N305, 2× i226-IT)	1
B026	Novanta Everest S NET drive	1 (first article)
B025	Elmo Platinum Twitter R70/100 class drive	1
B027	Drive carrier PCB	3 (spin 1)
B042	Safety MCU board (start on the TI LAUNCHXL2-570LC43)	1
B004	CubeMars RI85-PH frameless motor (test motor in a printed fixture)	1
B017	AksIM-2 encoder + ring	1
B078/B079	EtherCAT cable, J-LINK connectors, XT60, JST GH	—

C Tools

Oscilloscope (4-ch), bench supply ≥ 70 V current-limited, DC load, ESD bench, hot-air station, microscope, Ethernet tap or spare NIC for Wireshark (EtherCAT dissector).

D Manufacturing: drive carrier PCB (custom)

Specification (docs/05 §6): - 4-layer, 2 oz copper on power layers, ENIG finish; - the drive module footprint per the Novanta/Elmo integration manual; - 2 × 100 µF 100 V polymer + 1 µF X7R at the bus input; SMCJ75A TVS; - EtherCAT magnetics (integrated per the drive's

reference design) to two J-LINK connectors; - 2 × RS-422 receivers (e.g. THVD1450-class) converting BiSS-C MA/SLO to the 3.3 V single-ended inputs on Everest [R2]; - 5 V / 300 mA LDO for the encoder; - NTC dividers; STO input conditioning; an ADS1262 footprint (populated for KH-25/20 only).

Order 3 boards first, assembled by JLCPCB/PCBWay (BOM + centroid from your PCB tool).

E Assembly / setup steps

1. **RT computer OS:** Ubuntu 24.04 LTS, install a PREEMPT_RT kernel (6.8+ RT). Boot parameters: `isolcpus=2,3 nohz_full=2,3 rcu_nocbs=2,3 irqaffinity=0,1`. Disable C-states deeper than C1 in the BIOS.
2. Build **IgH EtherLab 1.6** with the native `igc` driver for the i226 (`--enable-igc`). Bind port 1 (EtherCAT) to the igc driver; leave port 2 on the normal stack for now (later it closes the ring).
3. Run `cyclictest -m -p95 -t2 -a2,3 -i1000 -D 1h` . **Target: max latency < 50 µs.**
4. Drive configuration: connect via the vendor tool (Novanta MotionLab / Elmo EASII). - Set the motor parameters from the datasheet (RI85-PH: 14 pole pairs, Kt 0.13 N·m/A, R_LL 0.1385 Ω [ds R1]). - Set the encoder to BiSS-C, 19 bit; current limit 5 A to start; I²t on; STO enabled.
5. Commutation: run auto-phasing with the rotor free. Check the electrical angle offset against a manual alignment.
6. Tune the current loop (vendor autotune), then verify with a step response.
7. **Safety MCU:** flash the monitor firmware (heartbeat watchdog on CAN from the RT computer, e-stop input reading, two relay-driver outputs, IMU tilt). Unit-test each trip path.

F Electrical

- Bench bus 48 V from the supply (current limit 3 A) → disconnect (enable = safety relay) → carrier XT60.
- STO pair from the safety relay → J-LINK pins 5/6/7 (signal level per the drive datasheet, **verify**).
- Encoder: JST GH 6-pin (5 V, GND, MA+, MA−, SLO+, SLO−), shielded, shield to the carrier chassis pin.
- First power-up procedure: 1. Inspect the board under a microscope. 2. Check the bus for shorts (> 10 kΩ). 3. Power at 12 V with a 0.2 A limit and check rails. 4. Raise to 48 V with a 0.5 A limit. 5. Only then raise the limit to 3 A.

G Testing (gate G0)

Test	Pass
EtherCAT: master sees the drive, OP state, 1 kHz with distributed clocks for 1 h	0 lost frames; DC sync error < 1 µs
Cyclic synchronous torque mode: ±0.5 N·m square wave on the free motor	current tracks with < 1 ms rise; no faults

Test	Pass
STO from each e-stop device while the motor runs	phase current → 0 in < 10 ms
Heartbeat loss (kill the RT process)	safety MCU opens the relay within 5 ms of the 2 ms timeout
Over-voltage: raise the supply to 72 V with the clamp disconnected (dummy load)	drive faults safely; clamp test: with the clamp connected the bus stays ≤ 72 V during a regen burst
Thermal: NTC reading matches a thermocouple	±2 °C

P03 KA-90P and KA-110P planetary actuators (first articles, then series)

Gate G1

A Visual reference

- Architecture of the KA family (same layout as the KA-130K cutaway, smaller): `renders/18_KA130K_cutaway.png` , `renders/19_KA130K_exploded.png`
- Envelope plots: `docs/img/envelope_KA-90P.png` , `envelope_KA-110P.png`
- Section (axial stack, output at top):

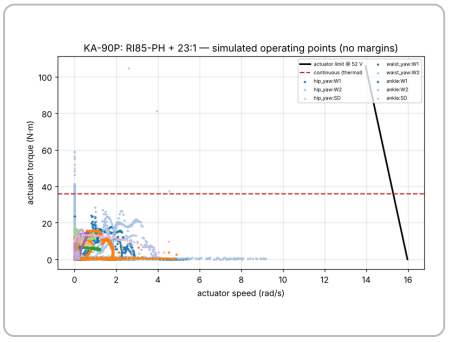
```
output flange ─┬─ crossed-roller bearing — stage-2 carrier (output)
               │  stage-2 ring (fixed in front housing) · 3–4 planets · sun
               │  stage-1 carrier → stage-2 sun ;  stage-1 ring · planets · sun (on rotor hub)
front housing ─┤
stator (bonded in finned housing) ── rotor (magnets on hub) ── rotor bearings ×2
motor encoder ring · annular drive carrier PCB · rear cap ── hollow Ø15 cable bore through everythin
```



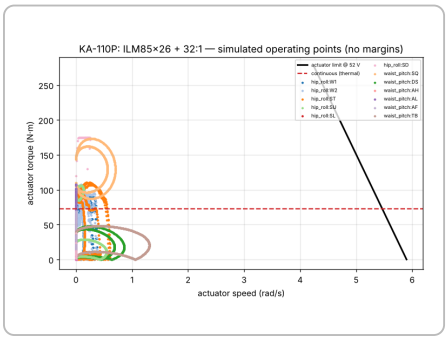
RENDER Architecture of the KA family (same layout as the KA-130K cutaway, smaller)



RENDER Architecture of the KA family (same layout as the KA-130K cutaway, smaller)



PLOT Envelope plots



PLOT Envelope plots

B Parts list (per actuator)

BOM	Item	KA-90P	KA-110P
B004 / B003	Frameless motor kit: CubeMars RI85-PH / TQ ILM85x26	1	1
B009 / B008	Planetary gear set 23:1 / 32:1 (sun, planets, ring ×2 stages, planet pins, needle rollers)	1	1
B014	Crossed-roller output bearing ~ID60/OD90	1	1
B016	Rotor bearings (thin-section deep-groove) + wave spring	2	2
B017	AksIM-2 output encoder (ring on carrier, readhead on housing)	1	1
B018 / B017	Motor encoder: MT6835 (KA-90P) / AksIM-2 (KA-110P)	1	1
B020	NTC B57861S	2	2
B022	Housing set: rear cap, motor housing (finned), front housing, rotor hub, carriers, output flange	1	1
B026 + B027	Everest S NET drive + carrier PCB	1	1
B081	M4 12.9 screws (output 8×/10×), M3 housing screws, Ø4 dowels, O-rings (front Ø92×2, rear Ø86×2), V-ring output seal, helicoils	set	set
—	Grease (Klüber Isoflex Topas class) ~8 g; thermal epoxy (≥ 1.5 W/m·K) ~10 g; Loctite 243, 638	—	—

C Tools

Arbor press with bearing-installation tools (flat drifts on the correct ring only), induction or oven heater (housing to 90 °C for stator fit), magnetic-safe non-ferrous tweezers, torque screwdrivers (0.5–6 N·m), dial indicator + magnetic base, feeler gauges, dyno (P01).

D Manufacturing

- Housings (B022): from the KA-130K STEP profiles scaled to the KA-90/110 interface table (docs/09 §4).
- 7075-T6; hard anodise black with masks on the bearing seats, the stator bore and the bonding pads.
- Machine the bearing seats **after** anodise if not masked.
- Critical tolerances: crossed-roller seat H6, rotor bearing seats H6/j5, stator bore +0.02/+0.05 (potting gap), ring-gear spigot H7/js6, coaxiality of all seats ≤ 0.01 mm.
- Gears: order from a gear shop to the table in docs/09 §5. Request a gear inspection report. Inspect on arrival: tooth count, OD, double-flank roll if available, hardness spot check (file test on a non-functional face).

E Assembly (KA-90P; KA-110P identical procedure)

1. **Clean** all parts (IPA). Lay them out on a lint-free mat in the order below.
2. **Stator into the motor housing.** - Heat the housing to 90 °C. - Apply thermal epoxy thinly to the bore. - Slide the stator in with the lead exit aligned to the cable slot. - Keep it pressed against the shoulder while it cures (24 h at room temperature, or as the epoxy data sheet says). - Pot the end turns with the same epoxy, leaving the rotor clearance free (check the bore with a gauge pin).
3. **NTCs:** glue two NTCs into the end turns at 180° (thin epoxy). Route the leads with the phase leads.
4. **Rotor hub:** - Press the two rotor bearings onto the hub (inner ring only) with Loctite 638. - Fit the stage-1 sun (keyed/pressed) and the motor encoder magnet/ring. - **Magnet safety:** the rotor is strongly attracted to the stator. Insert it with an axial guide rod through the bore and a printed sleeve. Never by hand.
5. **Insert the rotor** into the stator on the guide rod. Check free rotation and cogging feel. Measure the radial gap with plastic shim at 4 points; all must be within ±0.05 mm.
6. **Stage 1:** grease the ring gear and planets. Install the planets on the stage-1 carrier pins (needle rollers greased). Mesh them into the ring and onto the sun. Check for free rotation without tight spots.
7. **Stage 2:** fit the stage-2 ring into the front housing (spigot H7/js6, 2 dowels, Loctite 243 on the M3 screws at 1.2 N·m). Fit the stage-2 sun on the stage-1 carrier output. Install the planets on the output carrier.
8. **Crossed-roller bearing:** press onto the output carrier and into the front housing per the bearing maker's instructions. Tighten the clamp ring screws in a star pattern to the specified torque.
9. **Output encoder:** fit the AksIM-2 ring to the output carrier and the readhead to the housing. Set the readhead gap with the supplied shim and check the signal LED or the RLS tool.
10. **Close up:** join the front and motor housings (O-ring lightly greased, 6 × M3 at 1.2 N·m in a star). Fit the drive carrier PCB on the rear standoffs. Connect the phases (crimped ring lugs, M3 at 0.6 N·m), the encoders (JST GH) and the NTCs. Fit the rear cap.
11. **Output seal:** V-ring on the output flange. Fit the flange (dowels + 8 × M4 12.9 at **4.5 N·m** with Loctite 243).
12. **Label** the actuator serial and record the stator/rotor/gear serials in the log.

F Electrical

- Phases U/V/W to the drive (any order, then fix by auto-phasing). Twist the three leads.
- Bus via XT30 (16 AWG). EtherCAT + STO via the J-LINK pair.
- First power: 48 V, 3 A limit, STO enabled, drive in torque mode at 10 % limit.

G Testing on the dyno (gate G1, per actuator)

Test	Method	Pass (KA-90P / KA-110P)
Phase resistance, insulation	4-wire milliohm; 500 V insulation tester phases→housing	R_LL within ±10 % of datasheet; > 100 MΩ
Back-EMF constant	back-drive at 30 rpm output, scope a line-line voltage	Ke within ±5 % of datasheet
Kt at the output	current steps 10–100 % peak vs transducer torque	slope = Kt·N·η within ±8 % of model (calc table)
Efficiency map	speed × torque grid, forward and back-driven	η ≥ 0.88 at rated (model assumes 0.92; feed the measurement back into `ETA`)
Peak torque	2 s pulses to the drive limit	≥ 106 N·m / 276 N·m without slip or noise
Backlash, stiffness	lock the output, ±torque sweep, read both encoders	backlash ≤ 10 arcmin; record torsional stiffness
Back-drive torque	turn the output by hand through the transducer (drive off)	record (QDD target < 2 N·m)
Thermal step	hold the continuous torque (36 / 73 N·m) at 1 rad/s until steady	winding ≤ 120 °C at 25 °C ambient; record the time constant
Endurance	10^5 cycles of the walking torque profile (from `data/scenario_series.npz`)	no loss of torque, backlash growth < 2 arcmin, grease clean
E-stop	STO at full speed	stops; no damage

Record every result in `logs/actuators/<serial>.md`. **Only after the first article of each class passes, order and build the rest of that class.**

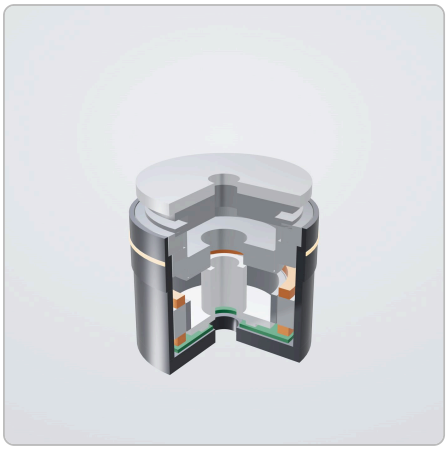
P04 KH strain-wave actuators (KH-25, KH-20, KH-14)

Gate G1

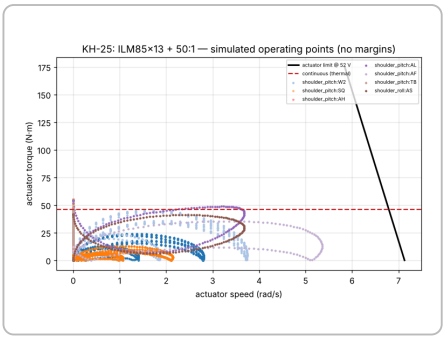
A Visual reference

- KH-25 cutaway: `renders/20_KH25_cutaway.png`
- STEP profiles: `cad/out/step/kh25__*.step`
- Envelope plots: `docs/img/envelope_KH-25.png`, `envelope_KH-20.png`, `envelope_KH-14.png`

```
output flange — torque flexure (Ti, 4 strain gauges) — crossed-roller — flexspline cup (output)
circular spline (fixed) — wave generator (elliptical plug + flexible bearing) on the hollow rotor hub
stator — rotor — motor encoder — drive carrier — rear cap — Ø12 cable bore
```



RENDER KH-25 cutaway



PLOT Envelope plots

B Parts list (per actuator)

BOM	Item	KH-25	KH-20	KH-14
B005 / B004 / B006	Motor: TQ ILM85x13 / RI85-PH / KM-50x20	1	1	1
B010 / B011 / B012	Leaderdrive LHSG size 25 / 20 / 14, 50:1 component set (WG, FS, CS)	1	1	1
B015	Crossed-roller output bearing	1	1	1
B016	Rotor bearings	2	2	2
B017, B018	AksIM-2 output + MT6835 motor encoder	1 + 1	1 + 1	1 + 1
B019	Torque flexure kit: 4 × HBK XY41 shear gauges, ADS1262 on the carrier	1	1	—
B023	Housing set incl. Ti flexure (KH-25/20)	1	1	1

BOM	Item	KH-25	KH-20	KH-14
B026 + B027	Everest S NET + carrier	1	1	1
—	Strain-wave grease (Harmonic Grease SK-1A class or the vendor's recommendation), gauge adhesive (M-Bond 610 class), protective coating (M-Coat)	—	—	—

C Tools

As P03, plus a strain-gauge bonding kit (clamps, Teflon tape, curing oven 120–175 °C for M-Bond 610), a fine soldering iron, a gauge installation tester.

D Manufacturing

- **Torque flexure (KH-25/20):** Ti-6Al-4V, 4 radial spokes (cross-section sized for 200 με at the repeated-peak torque: 127 N·m KH-25, 73 N·m KH-20; check by FEA). Gauge pads ground flat, Ra ≤ 0.8.
- **Housings:** tolerances per the Leaderdrive/Harmonic installation drawing. These are critical:
- circular spline and wave generator concentricity ≤ 0.01 mm;
- perpendicularity of the mounting faces ≤ 0.01 mm;
- the flexspline must have axial room to "breathe".

E Assembly

1. **Strain gauges first:** - Bond 4 shear gauges on the flexure spokes at ±45° to form a full Wheatstone bridge. The bridge is insensitive to bending and temperature when the gauges are matched. - Cure per the adhesive data sheet, wire to the 4-pin flexure connector and coat. - Measure the bridge: 350 Ω ± 0.5 %, insulation > 10 GΩ.
2. Motor stator and rotor as in P03 steps 2–5.
3. **Wave generator onto the rotor hub** (keyed, Loctite 638).
4. **Circular spline into the front housing** (dowelled, screws to spec). **Flexspline:** bolt the cup bottom to the crossed-roller inner ring/output carrier through the flexure, then torque to the vendor's value.
5. **Mesh:** grease the flexspline teeth and the wave-generator bearing per the vendor's quantity. Insert the wave generator into the flexspline at the vendor's specified orientation. Check the mesh by rotating: it must be smooth, with **no "dedoidal" (off-by-one-tooth) assembly**. Follow the vendor's alignment check exactly, because an incorrect mesh destroys the gear.
6. AksIM-2 ring on the output, readhead on the housing; motor encoder at the rear.
7. Close, seal and flange as P03 (KH-25 output 8 × M4 at 4.5 N·m; KH-20 8 × M3 at 2.0 N·m; KH-14 6 × M3).

F Electrical

As P03, plus the flexure bridge → ADS1262 on the carrier (excitation 5 V, gain 32, 1 kSPS, sinc3 filter).

G Testing (gate G1)

Test	Pass
Kt at the output, efficiency (η model 0.75)	measured η recorded; feed it back into `calc`
Firmware torque limit set to the gear's repeated-peak rating (127 / 73 / 23 N·m) and verified on the dyno	limit holds within ±5 %
Torque flexure calibration: ±full scale with lever + weights	linearity ≤ 1 % FS, hysteresis ≤ 0.5 % FS
Backlash / lost motion (dual encoder)	≤ 1 arcmin (strain-wave)
Collision detection: strike a padded pendulum into the output link at 1 rad/s	flexure detects within 5 ms; controller backs off
Thermal: continuous torque (46 / 64 / 9 N·m) at 2 rad/s to steady state	winding ≤ 120 °C
Endurance: 10^5 reach cycles at the arm torque profile	no ratcheting, lost motion growth < 0.5 arcmin

P05 KM-130 custom motors and KA-130K / KA-130H actuators

Gate G1

The custom motor (magnetic design, laminations, winding, magnets, rotor) is documented step by step in docs/13 §3. This phase assembles the actuator around it.

A Visual reference

- Cutaway and exploded views: `renders/18_KA130K_cutaway.png`, `renders/19_KA130K_exploded.png`
- Drawing sheets: `cad/out/drawings/ka130k__front_housing.svg`, `ka130k__motor_housing_finned.svg`, `ka130k__rotor_hub.svg`, `ka130k__s2_ring_gear_z54_m1.25.svg`, `ka130k__s2_carrier_output.svg`
- STEP: `cad/out/step/ka130k__*.step`
- Motor design sheet: `docs/generated/km_motor_designs.md`
- Envelope: `docs/img/envelope_KA-130K.png`

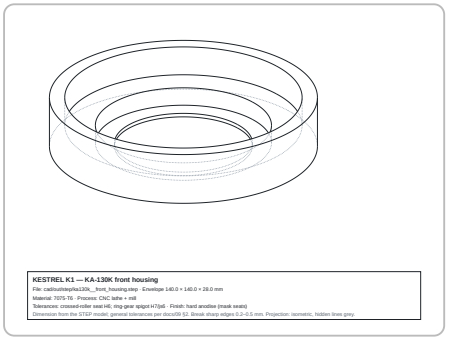
Axial stack, 0 → 124 mm: rear cap (fins) · annular drive PCB with GaN stage · motor encoder ring · motor housing with KM-130 stator (potted) · rotor with magnets + hollow hub carrying stage-1 sun · stage 1 nested inside the rotor bore (3 planets) · front housing with stage-2 ring · stage 2 (4 planets) · output carrier · crossed-roller ID90/OD120 · output flange · Ø20 cable bore throughout.



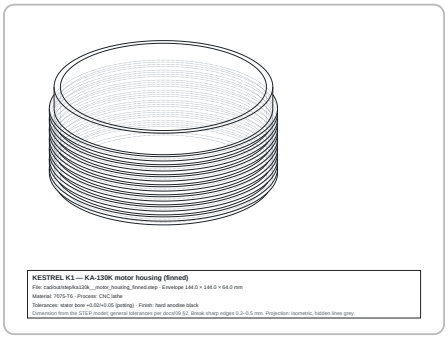
RENDER Cutaway and exploded views



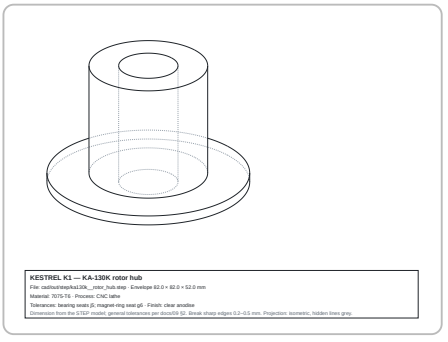
RENDER Cutaway and exploded views



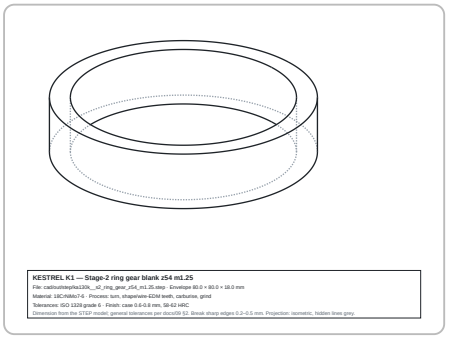
DRAWING ka130k / front housing



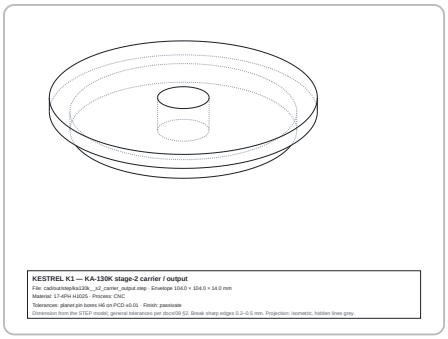
DRAWING ka130k / motor housing finned



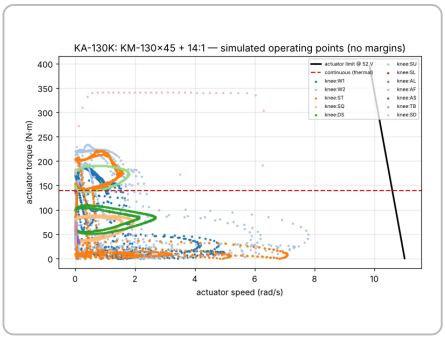
DRAWING ka130k / rotor hub



DRAWING ka130k / s2 ring gear z54 m1.25



DRAWING ka130k / s2 carrier output



PLOT Envelope

B Parts list (per actuator; build 2 × K and 2 × H)

BOM	Item	Qty
B001 (K) / B002 (H)	KM-130x45 / KM-130x40 motor (stator stack wound + rotor with magnets), see docs/13	1
B007	Planetary set 14:1 (stage 1 m0.8: sun z20, 3 × z15, ring z50; stage 2 m1.25: sun z18, 4 × z18, ring z54; pins + needle rollers)	1
B013	Crossed-roller ID90/OD120 class	1
B016	Rotor bearings 6808-class + wave spring	2
B017	AksIM-2 × 2 (motor and output)	2
B020	NTC	2
B021	Housing set: rear cap, motor housing, front housing, rotor hub, stage-1 carrier, stage-2 carrier/output, output flange	1
B025 + B027	High-current drive + annular carrier PCB	1
B081	12 × M5 12.9 (output), 12 × M5 (base), M3/M4 housing screws, 2 × Ø6 dowels each flange, O-rings, V-ring, helicoils M5	set
B080	thigh duct blower (in P07)	—

C Tools

As P03, plus a 2 t arbor press, an oven (housings to 100 °C), the magnet-safe rotor insertion jig (printed guide + Ø20 guide rod, see docs/13 §3.7), a hi-pot tester, an LCR meter.

D Manufacturing

- Housings: CNC 7075-T6, tolerances per the drawing sheets. Fins are turned. The stator bore is +0.02/+0.05 mm for the potting film.
- Gears: gear shop, 18CrNiMo7-6 carburised and ground (docs/09 §5). Stage-2 sun Lewis SF 2.7 at the momentary case (structure.py).
- Motor: docs/13.

E Assembly

1. **Motor first article test before building the actuator** (docs/13 §3.10): R, L, back-EMF, hi-pot, locked-rotor torque.
2. **Stator potting:** - Heat the motor housing to 100 °C. - Coat the bore with thermal epoxy (≥ 1.5 W/m·K). - Insert the stator with the phase leads at the cable slot. - Clamp axially and cure. - Pot the end turns, keeping a 1.5 mm clearance to the rotor path (check with a mandrel).
3. Install 2 NTCs in the end turns (one hot-spot side near the cable exit).
4. **Rotor:** - Press the bearings onto the hub; fit the stage-1 sun, the motor encoder ring and the wave spring. - Insert into the stator **with the jig**. The magnetic pull is several hundred newtons. Keep fingers clear.
5. **Stage 1** inside the rotor bore: ring z50 fixed to the housing spider; planets on the stage-1 carrier. Grease. Check free rotation by turning the rotor.
6. **Front housing with the stage-2 ring** (spigot H7/js6, dowels, M4 at 2.5 N·m), stage-2 sun on the stage-1 carrier, 4 planets on the output carrier.
7. **Crossed-roller** onto the output carrier, then into the front housing. Clamp ring screws in a star pattern to spec. Check output runout ≤ 0.02 mm with a dial indicator.
8. **Output AksIM-2 ring** on the carrier, readhead set with its shim.
9. **Close:** O-rings, M4 housing screws at 2.5 N·m in a star. Annular PCB on the rear cap (thermal pad between the GaN stage and the cap). Phases → PCB with crimped lugs. Rear cap on.
10. **Output flange:** 2 × Ø6 dowels, 12 × M5 12.9 at **8.5 N·m** (Loctite 243). This joint is also the overload slip fuse (SF 2.2 at the momentary torque); do not over-torque it.

F Electrical

- Bus through XT60 (12 AWG zone feed). J-LINK IN/OUT.
- Drive setup: KM-130x45 parameters from the **measured** motor test (not the model): pole pairs 21, Kt, R, L.
- Current limits: start at 30 A, raise to 99 A only on the dyno after the thermal test.
- Field weakening off until G1 completes.

G Testing (gate G1)

Test	Pass
Kt at the output vs model (model 0.310 × 14 × 0.92 = 4.0 N·m/A rms)	within ±8 %; update `calc` if outside
Peak torque 2 s	≥ 395 N·m at 99 A without slip or tooth noise
Torque-speed: points from `docs/img/envelope_KA-130K.png` with ×1.2 / ×1.15 margin at 52 V supply	all reachable
Thermal step test (critical, the knee's continuous margin is only 1.03): hold 117 N·m at 2 rad/s with the duct blower at its design flow until steady state	winding ≤ 120 °C at 25 °C ambient. **If it fails:** liquid-cooled jacket (docs/13 §6) or a 50 mm stack

Test	Pass
Walking duty cycle replay (W1/W2 knee torque series ×1.0) for 30 min	winding ≤ 110 °C, no drift in Kt
Backlash / stiffness	backlash ≤ 10 arcmin; stiffness recorded
Endurance 2×10^5 walking cycles	backlash growth < 2 arcmin; gear inspection clean
Overload: drive the output into a locked fixture at 1.5× momentary for 0.1 s (slip test)	the flange slips before any gear damage, or holds without damage; inspect

P06 Battery pack 16S6P with BMS and solid-state disconnect

Gate G0 (pack)

Hazard: a 1.68 kWh lithium-ion pack can deliver ~2 kA into a short and can burn violently. Build it in the battery area (P00), with insulated tools, never alone, and never leave charging unattended.

A Visual reference

- Layout and series path: `diagrams/battery_pack.svg`
- Power architecture: `diagrams/power_architecture.svg`
- Torso placement: `renders/17_torso_ghost.png`

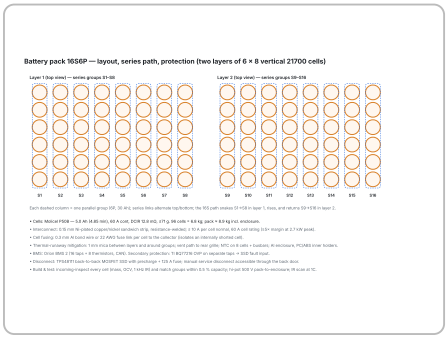


DIAGRAM Layout and series path

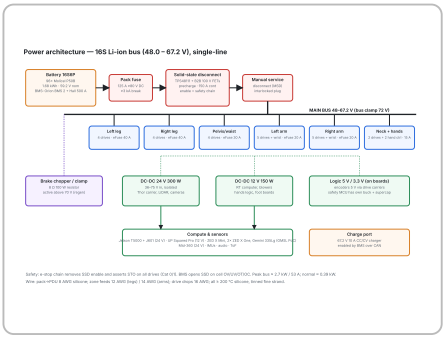


DIAGRAM Power architecture



RENDER Torso placement

B Parts list

BOM	Item	Qty	Spec
B028	Molicel INR-21700-P50B	96 + 10 spare	5.0 Ah, 60 A, ≤ 71 g [ds R3]; buy from an authorised distributor with lot-numbered invoices
B029	Orion BMS 2 (24-cell) + 500 A Hall sensor + cell-tap harness	1	16 taps + 8 thermistor inputs used
—	TI BQ77216 secondary over-voltage protector board	1	on separate taps
B030	Solid-state disconnect + precharge board (TPS48111, 2 × 8 × 100 V MOSFET, 20 Ω/10 W precharge)	1	150 A continuous
B031	Pack fuse 125 A, ≥ 80 V DC, ≥ 3 kA breaking	1 (+1 spare)	verify the DC rating on the datasheet before purchase

BOM	Item	Qty	Spec
B032	Enclosure (6061, machined), mica sheets 1 mm, PC/ABS cell holders (21.3 mm, 2-piece, 6 × 8), nickel-plated copper strip 0.15 × 8 mm + pure nickel 0.15 mm, 0.3 mm Al bond wire, 10 kΩ NTC ×10, 8 AWG silicone, SurLok Plus 5.7 mm pair, M5 busbar studs, Kapton tape, fish paper, foam-free compression pads	1 set	
—	Manual service disconnect (keyed plug, pilot contact)	1	

C Tools

Battery spot welder (kWeld kit with a ≥ 1500 A source, or a laser-weld service), insulated hand tools, 4-wire IR meter (1 kHz) or a battery tester, precision scale (0.01 g), DC load + programmable charger (or the robot charger), thermal camera, hi-pot / insulation tester, fire-safe workspace.

D Manufacturing

- **Enclosure:** 6061-T6 CNC, 2.5 mm walls, rear rails, vent path to the rear grille. M5 internal standoffs. Clear anodise. Line every inner face with 1 mm mica.
- **Cell holders:** two-layer 6 × 8 printed or moulded PC/ABS (do not use PLA, it softens at 55 °C).
- **Busbars:** cut nickel-plated copper strips into series links, pre-bend into a U across each series junction.

E Assembly

1. **Incoming inspection, 100 % of cells:** mass (reject > 71.0 g or < 68.5 g), OCV (all within ±20 mV), 1 kHz IR (reject outliers > 10 % above the median). Log by serial. Reject any wrap damage.
2. **Matching:** sort the 96 cells into 16 groups of 6 with equal summed capacity. Use IR and OCV as proxies, or a capacity test at 1C on a sample.
3. **Layer 1 (S1–S8):** insert cells into the holder, polarity alternating per series group per the diagram. Check polarity twice with a meter before any welding.
4. **Weld:** - Parallel collectors first (pure nickel on the cell, copper strip over it), then series links. - 3–4 welds per cell contact. Test-weld on a sacrificial cell and peel-test it (the weld nugget should tear the nickel). - Fit the **per-cell fuse link** (Al bond wire or a 22 AWG fuse link) between each cell and the collector.
5. **Layer 2 (S9–S16):** same. Fit the mica sheet between layers. Join layer 1 to layer 2 at S8→S9 with a doubled copper link.
6. **Cell taps:** solder each group's tap wire (22 AWG silicone, fused 1 A at the BMS end) to the group collector, in order B0…B16. **Verify the tap voltages ascend in 3.6 V steps before plugging into the BMS.**
7. **Thermistors:** 8 on cells (centre, corners, near the busbars) + 2 on the main busbars, held with Kapton and thermal paste.

8. **Main path:** B16+ → fuse → disconnect board → SurLok (+); B0− → Hall sensor → service disconnect → SurLok (−).
9. Fit the BMS and the disconnect board on the enclosure lid (the MOSFET array bolted to the aluminium for heat sinking). Close with foam-free compression pads holding the stack.

F Electrical

- BMS settings (Orion utility):
- cell OV 4.20 V, UV 3.00 V, balance from 4.05 V at 10 mV delta;
- charge 0–45 °C, discharge −10–60 °C;
- DCL 150 A (derate below 3.3 V/cell and above 50 °C); CCL 30 A (ramps to 0 at 4.15 V);
- CAN 500 kbit/s to the RT computer.
- Disconnect enable = AND(safety relay channel, BMS discharge-enable, BQ77216 OK, service-disconnect pilot).
- Precharge: the TPS48111 precharge path must complete (bus within 95 % of pack) before the main FETs close. Timeout 1 s → fault.

G Testing (pack gate)

Test	Pass
Insulation, pack terminals → enclosure, 500 V DC	> 100 MΩ
Tap voltages / BMS readings	all 16 cells within 10 mV of a reference meter
Precharge into a 5 mF dummy bus	completes < 0.5 s, resistor < 60 °C
Disconnect short-circuit test into a 10 mΩ shunt (at reduced voltage first: 12 V supply, then pack)	trips < 5 µs (TPS48111 1.2 µs class); FETs undamaged
Discharge 1C (30 A) to 3.2 V/cell into the DC load	thermal camera max cell < 45 °C; group voltages within 30 mV
Charge CC/CV 10 A to 4.20 V/cell with BMS supervision	balancing works; no cell > 4.21 V
Capacity	≥ 1.60 kWh to 3.0 V/cell
BMS trips (simulate OV/UV/OT by tap override, a thermistor in hot water)	the disconnect opens, CAN reports the fault

P07 Legs: hip, thigh with knee four-bar, shank with ankle push-rods, foot

Gate G2

Build the **left leg completely and pass G2 before building the right leg.**

A Visual reference

- Hip + knee joint assemblies (ghost): `renders/09_hip_knee_joints_ghost.png`
- Ankle + foot (ghost): `renders/16_foot_ankle_ghost.png`
- Skeleton rear: `renders/11_skeleton_rear.png`
- Knee four-bar geometry and G(θ): `diagrams/knee_linkage.svg`, `data/knee_fourbar.json`
- Ankle push-rods: `diagrams/ankle_linkage.svg`
- Drawings: `cad/out/drawings/l_hip__yaw_roll_yoke.svg`, `l_hip__roll_pitch_bracket.svg`, `detail__thigh_plate.svg`, `detail__knee_hub.svg`, `l_shank__shank_upper_fitting.svg`, `detail__ankle_cross.svg`, `detail__foot_plate.svg`, `l_thigh__knee_pushrod.svg`

Leg chain (left), with the joint frame origins in the zero pose:

Joint	Actuator	Location
hip yaw	KA-90P	pelvis bowl, axis vertical, 75 mm above the hip centre (0, 105, 990 mm)
hip roll	KA-110P	axis x through the hip centre, actuator body at x = −118 mm (gluteal)
hip pitch	KA-130H	axis y through the hip centre (0, 105, 915), body spans y 61–149 mm
knee actuator	KA-130K	axis y at z = 755 (160 mm below the hip)
knee joint	—	axis y at z = 495
ankle actuators	2 × KA-90P	at z = 400 (A, lateral rod) and z = 300 (B, medial rod)
ankle cross	—	z = 75; sole at z = 0



RENDER Hip + knee joint assemblies (ghost)



RENDER Ankle + foot (ghost)



RENDER Skeleton rear

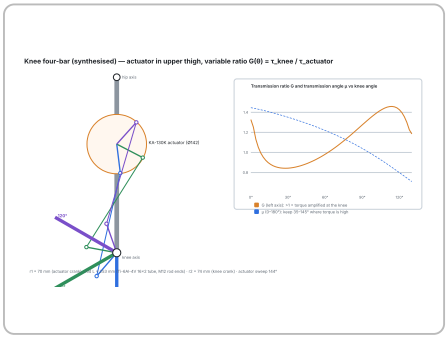


DIAGRAM Knee four-bar geometry and G(θ)

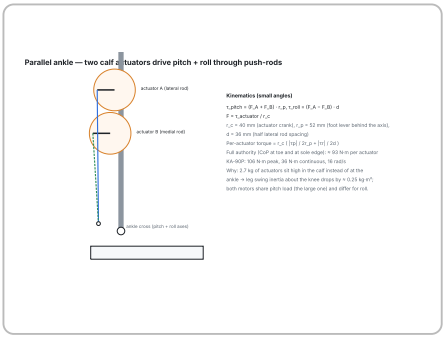
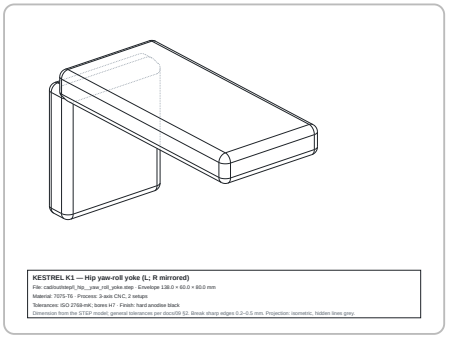
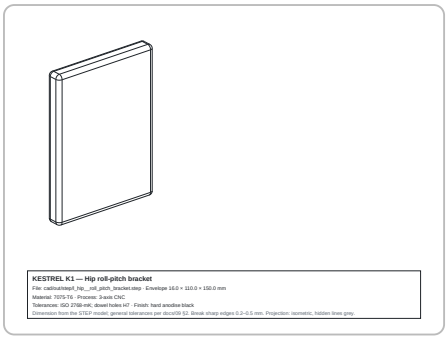


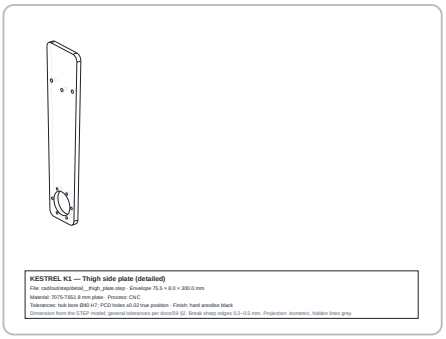
DIAGRAM Ankle push-rods



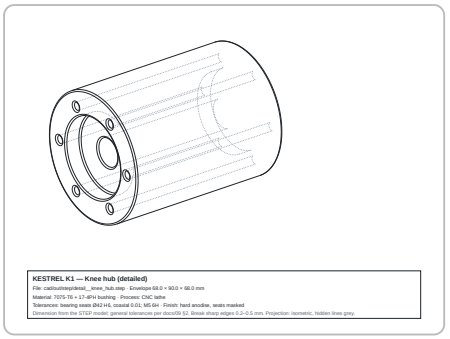
DRAWING l hip / yaw roll yoke



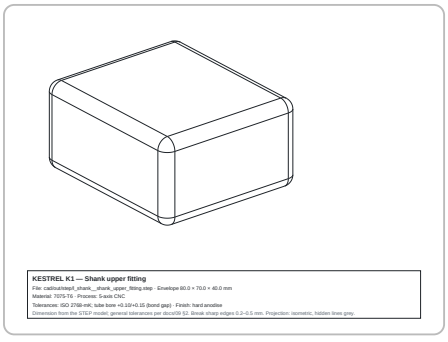
DRAWING l hip / roll pitch bracket



DRAWING detail / thigh plate



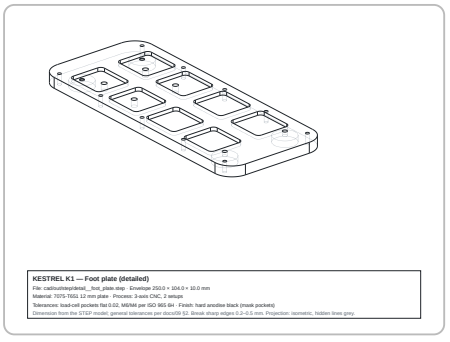
DRAWING detail / knee hub



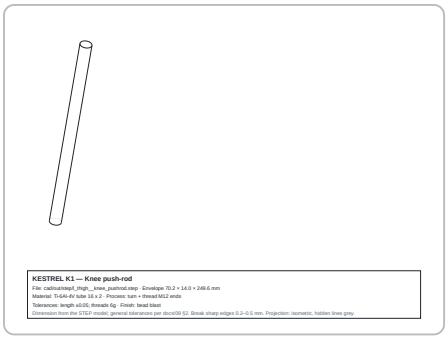
DRAWING l shank / shank upper fitting



DRAWING detail / ankle cross



DRAWING detail / foot plate



DRAWING l thigh / knee pushrod

B Parts list (one leg)

BOM	Item	Qty
(P03/P05)	KA-90P ×3 (hip yaw, ankle A, ankle B), KA-110P ×1, KA-130H ×1, KA-130K ×1, tested	6
B065	Yaw–roll yoke, roll–pitch bracket	1 + 1
B066	Thigh plates (medial + lateral), CF tube Ø40 × 2 × 150, end fittings	1 set
B067	Knee push-rod: Ti-6Al-4V tube 16 × 2, L = 253 mm (pin centres), 2 × M12 PTFE-lined rod ends (one LH thread for length adjustment)	1
B068	Actuator crank (r1 = 70 mm), knee crank (r2 = 74 mm at 229° in the shank frame), knee hub with 2 angular-contact bearings, 17-4PH pins	1 set
B069	Shank upper fitting, CF tube Ø40 × 2 × 230, lower fitting, 2 ankle push-rods (Ti 10 × 1.5) with M8 rod ends, 2 actuator cranks r_c = 40 mm	1 set
B070	Ankle cross (Ti), 4 × HK1212 needle rollers, 2 × Ø12 17-4PH pins, foot plate, foot levers (r_p = 52 mm, ±36 mm lateral), replaceable NBR sole	1 set
B051	4 × load cells + foot board	1 set
B077	Ankle TPU gaiter	1
B080	2 × 45 mm blowers + MJF duct for the thigh	1 set
B081	Fasteners: M5 12.9 (KA-130 flanges, 12 + 12), M4 12.9 (KA-110/90), M6 (foot clevis), M3 (shell bosses); Ø6/Ø4 dowels; DP490 structural epoxy	set

C Tools

Torque wrenches (2–25 N·m), dial indicators, height gauge, V-blocks, bonding fixture for the CF tubes (printed alignment jig), digital protractor (0.05°), leg stand (P01).

D Manufacturing

- **CF tube bonding (thigh and shank):** 1. Grit-blast the 7075 fitting bores (Ra 3–5 µm) and abrade the tube ends (P120). 2. Clean with IPA. 3. Apply DP490 with a 0.10–0.15 mm bond gap (glass-bead shims in the adhesive), 25 mm engagement. 4. Hold in the printed alignment jig so the fitting bores are coaxial within 0.05 mm and the clocking is correct. 5. Cure 24 h, then post-cure at 60 °C for 2 h. 6. Proof test one coupon per batch (docs/09 §9).
- **Knee push-rod:** 1. Ti tube, ends bored and tapped M12 (one RH, one LH) 18 mm deep. 2. Set the length to 253.0 ± 0.1 mm between rod-end ball centres. 3. Lock with jam nuts at 25 N·m.
- **Foot sole:** cast NBR/PU tread 10 mm in a printed mould, bonded to a 1 mm aluminium carrier screwed to the foot plate (12 × M4), so it can be replaced.

E Assembly (left leg)

1. **Hip yaw:** mount KA-90P in the pelvis bowl later (P08). For the leg stand, mount it on the stand carriage plate in the same pattern.
2. **Yaw–roll yoke** onto the hip-yaw output flange: 2 dowels, 8 × M4 12.9 at 4.5 N·m, Loctite 243.
3. **Hip roll actuator** (KA-110P) into the yoke at x = −118 mm, output pointing +x. Base flange 10 × M4 at 4.5 N·m.
4. **Roll–pitch bracket** onto the KA-110P output (dowels, 10 × M4). Check the roll axis passes through the hip centre: dial indicator on a test bar through the KA-130H seat, rotate the roll, runout ≤ 0.1 mm.
5. **Hip pitch actuator** (KA-130H) into the bracket, coaxial with the hip centre, output facing medially. Base 12 × M5 at 8.5 N·m.
6. **Thigh:** medial and lateral plates onto the KA-130H output flange (12 × M5 12.9 at 8.5 N·m, dowels). Fit the CF tube assembly between the plates. Mount the **KA-130K** at z = 755 (thigh frame: 160 mm below the hip axis), output on the lateral side.
7. **Knee hub** between the lower plate ends: angular-contact pair, back-to-back, with 50 N axial preload via a shim stack. The hub must turn with 0.2–0.4 N·m drag and no play.
8. **Shank:** shank upper fitting onto the knee hub (6 × M5 at 8.5 N·m). Mount both ankle actuators on the fitting brackets (A at z = 400 output lateral; B at z = 300 output medial).
9. **Knee four-bar, zero setting:**
10. Set the knee to 0° (straight) using the Ø4 zero pin through the knee hub and plate.
11. Command the KA-130K to its zero (output encoder), so the actuator crank sits at **φ0 = 277°** in the thigh frame (pointing almost straight down, slightly forward).
12. Fit the actuator crank (r1 = 70 mm) on the KA-130K output and the knee crank (r2 = 74 mm) on the knee hub at **229°** in the shank frame.
13. Install the push-rod and adjust its length (LH/RH thread) until both zero pins slide in freely. Torque the jam nuts.
14. **Check the knee ratio** at 0°, 40°, 80°, 120° by reading the actuator and knee encoders: the incremental ratio must match G(θ) in `data/knee_fourbar.json` within ±3 % (0.86 at 20–40°, 1.38 at 100°).
15. **Ankle:**

1. Ankle cross into the shank lower fitting (pitch pin) and the foot clevis (roll pin), needle rollers greased, pins m6 with retaining screws.
2. Fit the push-rods from the actuator cranks (r_c = 40 mm) to the foot levers (52 mm behind the axis, ±36 mm lateral).
3. Zero: foot flat on a surface plate, both ankle actuators at encoder zero, rod lengths adjusted until the foot is level within 0.2° in pitch and roll.

16. **Hard stops:** fit polyurethane bumpers at each joint end of range and check they contact 1–2° before the linkage limits.
17. **Thigh cooling duct:** fit the MJF duct over the KA-130K/H fins, 2 blowers (intake at the knee, exhaust under the pelvis), filter foam.
18. **Foot board** with 4 load cells in the foot plate pockets; sole carrier on top; ankle gaiter.

F Electrical

- Leg zone feed: 12 AWG pair from the PDU (eFuse 40 A) through the hip yaw bore. Daisy chain to the 6 drives with XT30 drops (16 AWG).
- EtherCAT order: hip yaw → hip roll → hip pitch → knee → ankle A → ankle B (J-LINK OUT → IN).
- Foot board (load cells + IMU) → ankle A drive's auxiliary I/O or its own EtherCAT slave (design choice; default: an SPI bridge on the ankle A carrier).
- Cable through each actuator's hollow bore with a 30 mm service loop each side, clipped every 80 mm.

G Testing (gate G2: leg on the stand)

Test	Pass
Joint zeroing: all six encoders vs zero pins	±0.05°
Range of motion, slow (5 °/s), each joint	full software range, no interference, bumpers contact as designed
Kinematics: foot pose vs forward kinematics at 20 poses, measured with a height gauge / tracker	≤ 1.5 mm
Knee ratio check	G(θ) within ±3 %
Gravity compensation, leg hanging	holds with < 10 % torque error vs the model
Loaded squat on the stand (counterweight sets the load to 45 kg, half the design mass), to 90° knee, 20 cycles	tracking error < 2°, temperatures logged, no noise
Single-leg stance loading on the stand: 93 kg equivalent for 60 s	KA-130K winding rise matches the dyno model within 10 %
Ankle: CoP sweep heel→toe at full load	ankle actuators within 106 N·m; load-cell CoP within ±5 mm of the target
Stair-step trajectory replay (`ST` scenario, ×0.5 speed, then ×1.0)	no limit violations, temperatures OK

Test	Pass
E-stop at each test	joints go limp, the stand catches the load

P08 Pelvis, waist, torso frame, power distribution and compute

Gate G2/G3

A Visual reference

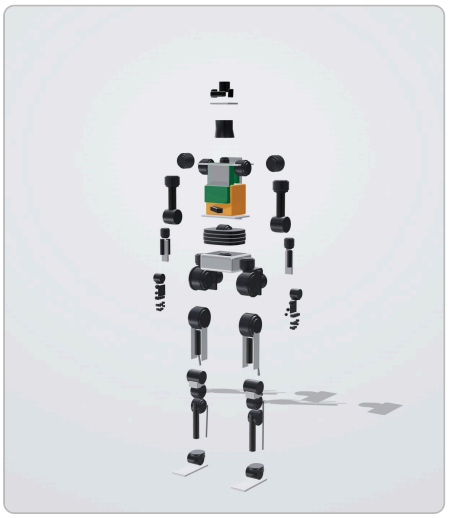
- Torso internals: `renders/17_torso_ghost.png`
- Skeleton: `renders/10_skeleton_front.png` , `renders/13_exploded_skeleton.png`
- Drawings: `cad/out/drawings/pelvis__pelvis_bowl.svg` , `torso__spine_plate.svg`
- Power and comms: `diagrams/power_architecture.svg` , `diagrams/comms_topology.svg`



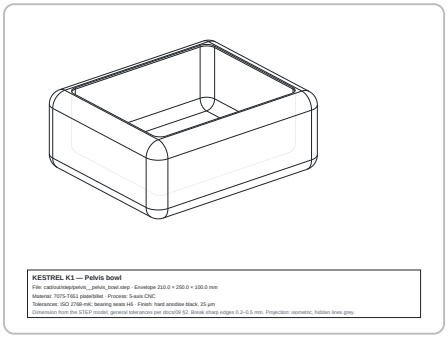
RENDER Torso internals



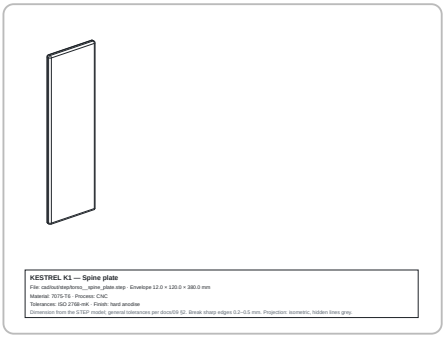
RENDER Skeleton



RENDER Skeleton



DRAWING pelvis / pelvis bowl



DRAWING torso / spine plate

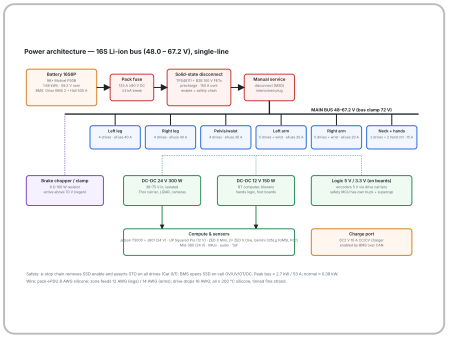


DIAGRAM Power and comms

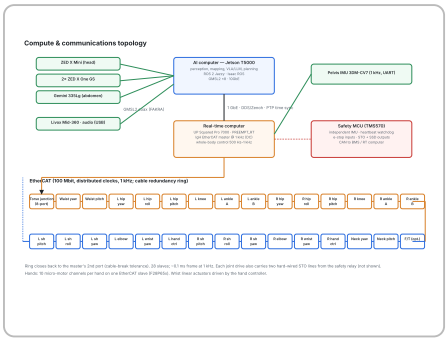


DIAGRAM Power and comms

B Parts list

BOM	Item	Qty
B063	Pelvis bowl (7075, 5-axis) with M10 lifting eye	1
(P03)	KA-90P (waist yaw) + KA-90P ×2 (hip yaw, already in the legs)	1
(P03)	KA-110P (waist pitch)	1
B064	Spine plate, upper/lower ribs, CF closeout panels, battery rails, lifting eye	1 set
(P04)	KH-25 shoulder-pitch actuators	2
(P04)	KH-14 neck-yaw actuator	1
B035, B036	PDU board (6 eFuse zones, brake chopper, telemetry), brake resistor	1 each
B033, B034	DC-DC 24 V 300 W, 12 V 150 W (36–75 V input)	1 each
B038–B040	Jetson T5000 + J601 carrier + NVMe, cold plate + heat pipes	1
B041	UP Squared Pro 7000 (RT computer)	1
B042, B050	Safety MCU board + IIM-46230	1
B049	MicroStrain 3DM-CV7 on a 4-point silicone isolator (pelvis)	1
B047	Gemini 335Lg (abdomen, 35° down)	1
B080	2 torso blowers + ducts + filters	1 set
(P06)	Battery pack	1
B077	Waist TPU bellows	1

C Tools

As P07, plus thermal paste and pads, crimp tools (XT60, SurLok, Gecko-MT, JST GH), a label printer.

D Manufacturing

- Pelvis bowl: 5-axis 7075 with all seats (hip yaw ×2, waist yaw) bored in one setup for relative position ≤ 0.02 mm. Hard anodise with the seats masked.
- Spine and ribs: CNC 7075 plates. CF closeouts cut from 2 mm CF sheet and bonded/screwed.

E Assembly

1. **Pelvis:** fit both hip-yaw KA-90P (outputs down) and the waist-yaw KA-90P (output up) into the bowl seats (8 × M4 at 4.5 N·m each). Mount the IMU isolator at (−60, 0, +50 mm) from the hip-centre midpoint, x-axis forward ±0.5°.

2. **Waist:** yoke onto the waist-yaw output; KA-110P (waist pitch) into the yoke with its axis y at z = 1085 mm.
3. **Torso frame:** spine plate onto the waist-pitch output flange (10 × M4 at 4.5 N·m), then the ribs. Battery rails inside. KH-25 shoulder-pitch actuators on the upper rib at y = ±135 mm, outputs outward, axes at z = 1420 mm. Neck-yaw KH-14 on top.
4. **Electronics bay:** - Jetson + J601 on the front of the upper ribcage with the cold plate facing forward into the intake duct. - RT computer and safety MCU on the rear of the spine plate (conduction pad). - PDU and DC-DCs beside the battery bay (front lower).
5. **Ducts:** torso intake under the chest light → Jetson cold plate → blowers → rear grille. Thermal paste on the cold plate, heat pipes clamped.
6. **Mount the legs** to the pelvis (hip-yaw outputs → yaw–roll yokes) with the robot hanging from the gantry lifting eyes.
7. **Battery:** slide into the rails from the back, latch, connect the SurLok pair to the PDU, then the service disconnect.

F Electrical

- PDU zone outputs:
- left leg / right leg: 40 A, 12 AWG;
- pelvis/waist: 30 A, 14 AWG (waist yaw, waist pitch, both hip yaws);
- left arm / right arm: 20 A, 14 AWG;
- neck + hands: 15 A, 16 AWG.
- 24 V rail → J601 (XT30, 12 AWG), Mid-360, GMSL camera power through the J601.
- 12 V rail → RT computer, blowers, hand logic, foot boards, safety MCU (its own fuse).
- EtherCAT: RT computer port 1 → torso junction → waist → left leg chain → right leg chain → arms → neck → back to port 2.
- E-stop chain: FORT receiver + back button + tether socket → safety relay (back of the spine) → STO bus (all J-LINK pins 5/6/7) + disconnect enable.
- CAN: BMS ↔ RT computer ↔ safety MCU (120 Ω terminations at the ends).
- Bond BUS– to the chassis at the PDU star point (10 mm² strap). Mask the anodise at the bonding pads.

G Testing (gate G2/G3 preparation)

Test	Pass
Insulation, bus → chassis with the bond lifted, 500 V	> 10 MΩ (then refit the bond)
Power-up sequence (pack → precharge → bus → DC-DCs → computers)	rails within ±3 %; precharge < 0.5 s
eFuse trip test per zone with an electronic load	trips at the set current ±10 %
EtherCAT: all slaves present, ring redundancy (pull one cable mid-run)	OP state maintained; 1 kHz DC, jitter logged < 20 µs

Test	Pass
Thermal: Jetson at 100 W synthetic load for 30 min in the closed torso	module below throttle temperature
Waist yaw/pitch range and holding with the arms mounted later	per P07 criteria
Safety MCU trip paths (tilt the torso > 35° on the gantry, kill the heartbeat)	Cat 1 sequence then STO

P09 Arms: shoulders, upper arms, elbows, forearms, wrists

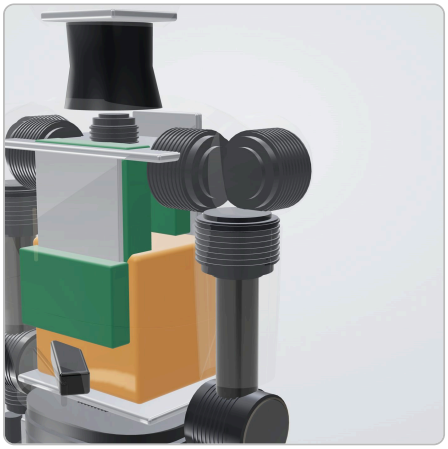
Gate G2

A Visual reference

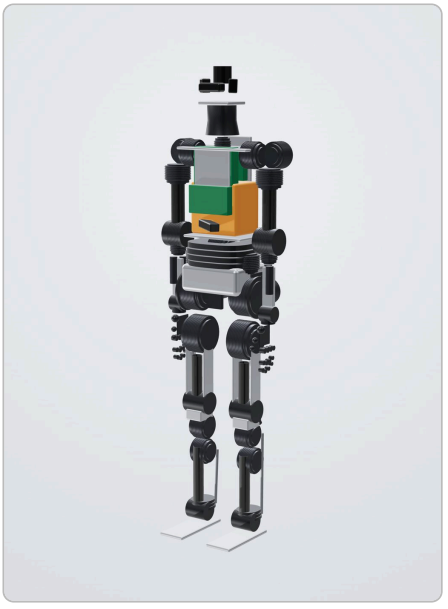
- Shoulder (ghost): `renders/15_shoulder_ghost.png`
- Skeleton: `renders/10_skeleton_front.png`
- KH-25 cutaway: `renders/20_KH25_cutaway.png`

Arm chain (left), zero pose (arm hanging, palm facing the thigh):

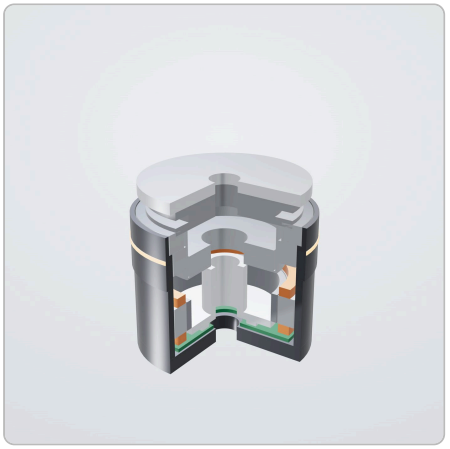
Joint	Actuator	Location
shoulder pitch	KH-25	in the chest, axis y at (0, 135, 1420)
shoulder roll	KH-25	axis x, body behind the shoulder centre at x = −45
shoulder yaw	KH-20	axis along the arm at z = 1340
elbow	KH-20	axis y at z = 1120
wrist yaw	KH-14	axis along the forearm at z = 1072
wrist pitch/roll	2 linear actuators	z = 985; wrist cross at z = 860



RENDER Shoulder (ghost)



RENDER Skeleton



RENDER KH-25 cutaway

B Parts list (one arm)

BOM	Item	Qty
(P04)	KH-25 (roll), KH-20 ×2 (yaw, elbow), KH-14 (wrist yaw), tested	4
B071	Shoulder bracket, roll-output bracket, upper-arm CF tube Ø50 × 2 + fittings, elbow-output bracket, forearm frame, Ti wrist cross + needle rollers	1 set

BOM	Item	Qty
B024	Wrist linear actuators (DCX 16 L + GPX 16 3.9:1 + KSS Ø6 × 2 ball screw + miniature guide), with M4 rod ends	2
B054	VL53L5CX ToF module (forearm)	1
—	Clock-spring cable cassettes for shoulder pitch (240° range) and roll	2

C Tools

As P07.

D Manufacturing

- Upper-arm CF tube bonding as in P07 §D.
- Wrist cross (Ti-6Al-4V): two perpendicular Ø6 H7 pin bores intersecting within 0.01 mm. Lever arms 22 mm (pitch) and 16 mm (roll) to the linear actuator rod ends.

E Assembly

1. **Shoulder bracket** onto the KH-25 shoulder-pitch output (8 × M4 at 4.5 N·m, dowels). Mount the KH-25 roll actuator on the bracket, axis x through the shoulder centre (check with a test bar as in P07 step 4).
2. Fit the **clock-spring cassette** at the pitch joint. Wind it to the middle of its travel at the zero pose.
3. **Roll-output bracket** → KH-20 shoulder-yaw actuator (axis along the arm) → upper-arm tube assembly → KH-20 elbow actuator at 300 mm (shoulder-to-elbow centre distance ±0.2 mm).
4. **Forearm:** elbow-output bracket → KH-14 wrist-yaw actuator → forearm frame carrying the two linear actuators and the hand controller (P10).
5. **Wrist:** wrist cross on its pins; linear actuator rod ends to the cross levers (A at +x/+y, B at −x/−y). Zero: wrist straight, both screws at mid-stroke, rods adjusted until the hand axis is coaxial with the forearm within 0.3°.
6. Bumpers at the elbow (0° and −145°) and the wrist limits.

F Electrical

Arm zone 14 AWG from the PDU (eFuse 20 A) through the shoulder-pitch bore. EtherCAT daisy chain: shoulder pitch → roll → yaw → elbow → wrist yaw → hand controller. Torque-flexure bridges are wired on each KH-25/KH-20 carrier.

G Testing (gate G2: arm on the bench, shoulder clamped)

Test	Pass
Range of motion (slow), all 7 joints	full range, no cable snag (watch the clock-springs)

Test	Pass
FK accuracy at the hand flange, 20 poses	≤ 2 mm
Payload: 5 kg at full forward reach for 60 s	shoulder winding rise within the model; no gear-limit warnings (shoulder ≤ 127 N·m)
Fast reach (AF scenario replay)	tracking error < 3°, speeds reach 6 rad/s shoulder / 7 rad/s elbow
Collision detection: padded impact on the forearm at 0.5 m/s	stops within 50 ms; impact force logged
Wrist torque: 13 N·m pitch / 10 N·m roll against a fixture	holds; screws not back-driven

P10 Hands (10 actuated DOF each)

Gate G2

Build **one finger first** (index: 3 actuators) and run its endurance test before building the rest.

A Visual reference

- Hand detail render: `renders/08_hand_detail.png`
- Actuation map: `diagrams/hand_architecture.svg`
- CAD: `cad/hand.py` (palm, phalanges, joints, internals: `hand_internals()`), STEP `cad/out/step/l_hand__palm_shell.step`
- Reference design: ILDA hand (Kim et al., Nat. Commun. 2021), the architecture this hand follows [R5].



RENDER Hand detail render

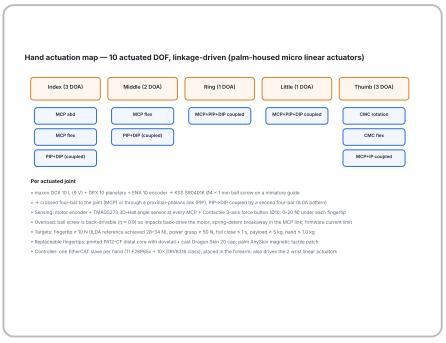


DIAGRAM Actuation map

B Parts list (one hand)

BOM	Item	Qty
B058	Micro linear actuator: maxon DCX 10 L 9 V + GPX 10 (16:1 for MCP flexors; 4:1 for thumb rotation and index abduction) + ENX 10 encoder + KSS SR0401K Ø4 × 1 mm ball screw + miniature guide (MGN5 class)	10
B059	TMAG5273 3D Hall angle sensors + Ø3 × 1 mm diametric N52 magnets	16
B052	Contactile 3D force buttons Ø10, 0–20 N	5
B053	Palm magnetic tactile skin (5 × MLX90393 + magnetised Dragon Skin 10 + iron microparticles, AnySkin pattern)	1
B060	Hand controller PCB (F28P65x + 10 × DRV8316-class + EtherCAT + 2 wrist-actuator channels), per TIDA-010992	1
B061	7075 palm frame, PA12-CF (MJF) phalanx links and four-bar links, Ti-6Al-4V Ø2 pins, 17-4PH bushings, cast Dragon Skin 20 fingertip caps, palm shell	1 set
B081	M1.6/M2 screws, E-clips, 0.2 mm PTFE washers, Loctite 222	set

C Tools

Stereo microscope, fine tweezers, M1.6 drivers, torque screwdriver 0.05–0.5 N·m, reamers Ø2 H7, small press for pins, silicone casting kit (vacuum chamber), printed moulds, force gauge 0–100 N, the finger endurance rig (printed frame + load spring + cycle counter).

D Manufacturing

- **Palm frame (7075):** CNC; holds the 10 actuators in two rows, guide rails and the MCP pivot bores (Ø2 H7). Hard anodise.
- **Phalanx and four-bar links:** MJF PA12-CF, reamed to Ø2 H7 after printing. Bushings are pressed in. Link lengths follow the ILDA crossed four-bar for MCP and PIP, with DIP coupled by a second four-bar. Size them in a parametric sketch so DIP ≈ 0.8 × PIP angle.
- **Fingertip caps:** cast Dragon Skin 20 (Shore 20A) in a two-part printed mould over a PA12-CF distal core with a dovetail. The Contactile button sits under a 1.5 mm silicone layer.
- **Palm skin:** mix Dragon Skin 10 with iron microparticles, cast 2 mm, magnetise in a pulse magnetiser or with an array of N52 blocks (AnySkin method [R5]).

E Assembly

1. **Actuator sub-assembly:** motor + gearhead + ball screw coupling (set screw on a flat, Loctite 222), nut carriage on the MGN5 guide, rod end. Check smooth travel; measure the no-load current over the full stroke (< 60 mA).
2. **Fingers:** assemble the phalanges with pins + PTFE washers + E-clips. Fit the four-bar links. Fit a magnet in each MCP and PIP pin end, and the TMAG5273 flex PCB in the facing link.
3. **Mount the actuators** in the palm frame. Connect each nut carriage to its MCP crank (or PIP link through the proximal phalanx for index/middle).
4. **Thumb:** - CMC rotation actuator (4:1, fast) turns the thumb base on a 2-bearing pivot. - CMC flexion actuator. - MCP+IP coupled flexion through a four-bar.
5. **Overload breakaway:** the spring-detent in each MCP crank must release at 1.5× the stall force (set with the force gauge) and re-seat by hand.
6. Fingertip caps (dovetail slide-on, one drop of silicone adhesive), palm skin (4 × M2 screws), palm shell.
7. **Wiring:** motor leads and encoder flex through the wrist on a 0.5 mm-pitch flat flex with a service loop to the controller in the forearm.

F Electrical

- Hand controller: 12 V from the neck/hands zone; EtherCAT IN/OUT.
- Each motor on a DRV8316-class channel with a current limit (stall-protect at 0.4 A for the DCX 10 L at 9 V).
- I²C buses: 2 × for the TMAG5273 sets (address-select via mux), 1 × for the Contactile buttons, 1 × for the palm skin.
- Wrist linear actuators on two higher-current channels (DCX 16 L).

G Testing (gate G2: hand)

Test	Pass
Each joint range and the finger angle sensors vs a protractor	±1°
Fingertip force, straight and bent (force gauge)	≥ 10 N (target; ILDA achieved 28–34 N [R5])
Power grasp on a 50 mm cylinder	≥ 50 N total
Close time, open → full fist	≤ 1.0 s
Payload: 5 kg handle hook grasp for 60 s	no slip, motor temperature < 60 °C
Back-drive: push a fingertip 20 N	finger yields (ball screw back-drives) without damage
Breakaway: overload the MCP	detent releases, re-seats
Endurance (first finger): 10^4 close/open cycles at 50 % load	no loss of force > 10 %; no link wear beyond 0.05 mm at the pins
Tactile: Contactile and palm skin respond to 0.1–20 N	monotonic, calibrated

P11 Neck, head, cameras, LiDAR, calibration

Gate G3

A Visual reference

- Head detail: `renders/07_head_detail.png`
- Sensor placement: docs/06
- CAD: head shell `cad/out/step/head__head_shell.step`; visor and side windows in `cad/build_robot.py`



RENDER Head detail

B Parts list

BOM	Item	Qty
(P04)	KH-14 neck pitch (neck yaw installed in P08)	1
B072	Neck yoke, head frame (6061 plate + PA12-CF bracketry)	1 set
B045	ZED X Mini (behind the visor, 10° down)	1
B046	ZED X One GS (temples, yaw ±55°)	2
B048	Livox Mid-360 (crown)	1
B055	ReSpeaker XVF3800 + speaker	1
B054	VL53L5CX (head rear)	1
B075	Visor (PC, smoked) + side windows + LiDAR window	1 set
—	GMSL2 FAKRA-Z coax, 1.2 m × 4; Ethernet (Mid-360); USB (mic)	—

C Tools

ChArUco calibration board (A1, rigid), AprilTag ground board, a laptop with Kalibr and the ZED SDK, a level, a lint-free optics kit.

D Manufacturing

- Head frame: a 6061 plate carrying the camera bracket (stiff: cameras must not move relative to each other, so first mode > 150 Hz), the LiDAR seat and the neck-pitch flange.
- Visor: thermoform 3 mm PC over a printed buck; cut the aperture windows; AR + hard coat on the camera apertures (outsourced optical coating, or buy coated PC windows and bond them in).

E Assembly

1. Neck yoke on the neck-yaw output; neck-pitch KH-14 in the yoke (axis y at z = 1555).
2. Head frame on the neck-pitch output (6 × M3 at 2.0 N·m, dowels).
3. Cameras on the rigid bracket: ZED X Mini centred 66 mm forward of the neck axis at z = 1655; the two ZED X One GS at the temples, yawed ±55°. Torque the camera screws per the manufacturer and add thread-locker.
4. Mid-360 on its seat, its top ≈ 2 mm below the crown window. Ethernet + 24 V. Mic array at the lower rear behind its grille.
5. Route the 4 coax cables and the Ethernet through the neck with a twist-capable service loop (neck yaw ±80°, pitch −35/+50°).
6. Fit the head shell (two halves) and the visor (gasketed). Clean the windows.

F Electrical

- GMSL2 coax to the J601 deserialiser ports 0–3. Power-over-coax from the J601.
- Mid-360: 24 V rail (9–27 V input [ds R4]), Ethernet to the Thor 10GbE switch port, PTP enabled.
- Hardware frame sync: all GMSL cameras on one J601 frame-sync line.

G Testing and calibration (part of gate G3)

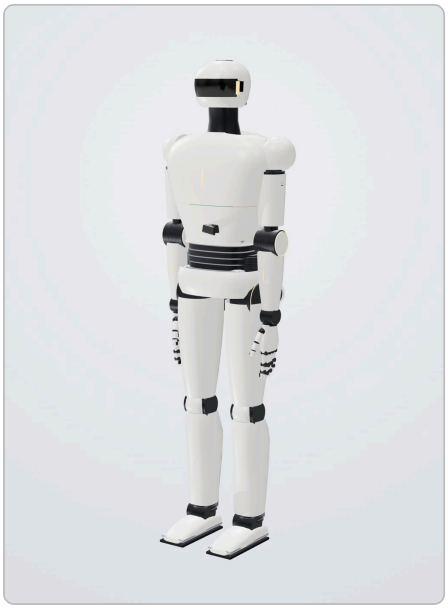
Test	Pass
All streams at rate (ZED X Mini 30–60 fps, One GS 30 fps, Gemini 30 fps, Mid-360 10 Hz) with PTP timestamps	no drops over 10 min; time offset < 1 ms
Stereo + intrinsics calibration (Kalibr / ZED tool)	reprojection RMS < 0.3 px
Camera–IMU extrinsics + time offset (Kalibr)	converged; time offset stable within 0.5 ms
LiDAR–head extrinsics	plane residual < 1 cm
Head–body kinematic calibration (AprilTag board, 30 poses)	≤ 2 mm / 0.2° at 1 m
Neck range with cables, 1,000 cycles	no intermittent camera link errors

P12 Harness, cooling, shells and final assembly

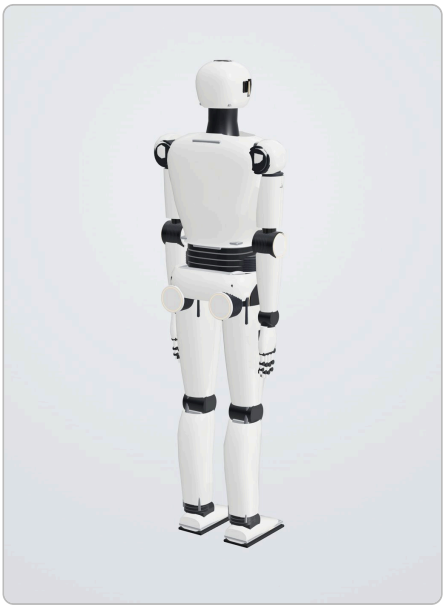
Gate G3

A Visual reference

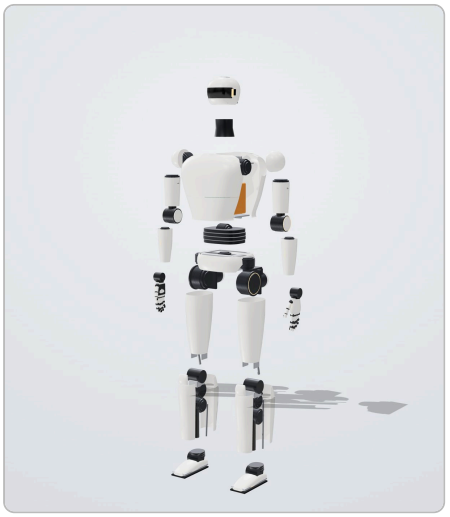
- Fully assembled: `renders/05_threequarter_front.png`, `renders/06_threequarter_rear.png`
- Exploded (shell panel splits): `renders/12_exploded.png`
- Ghost (shells over the structure): `renders/14_ghost_threequarter.png`
- Shell STEP files: `cad/out/step/*front*.step`, `*rear*.step`, `torso__chest_plate.step`, `torso__abdomen_plate.step`, `torso__back_door.step`, `pelvis__pelvis_*.step`, `l_shoulder__deltoid_cap.step`, `head__head_shell.step`, `l_foot__foot_shell.step`



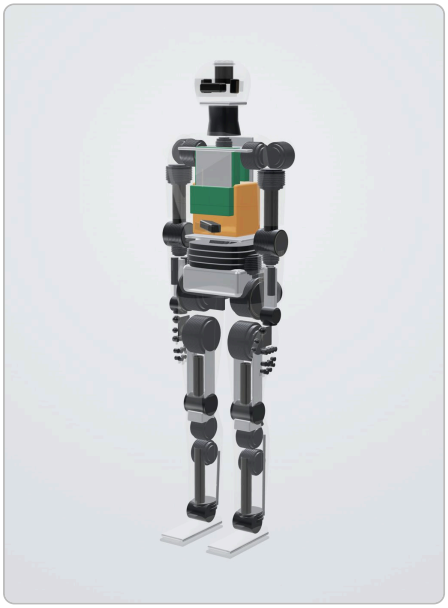
RENDER Fully assembled



RENDER Fully assembled



RENDER Exploded (shell panel splits)



RENDER Ghost (shells over the structure)

B Parts list

BOM	Item	Qty
B073	MJF PA12 shells (≈ 42 panels)	1 set
B074	Primer + 2K satin warm-white + matte graphite + clearcoat (shop)	1
B076	N52 disc magnets Ø10 × 3 + steel pads, captive M3 quarter-turn studs, M3 heat-set inserts, 3 mm silicone isolators	1 set
B077	Waist bellows, ankle gaiters	3
B078/B079	Harness wire, connectors, sleeving, heat-shrink, P-clips, labels	1 set
B080	Duct filters	6

C Tools

Heat-set insert iron, harness nail-board (printed 1:1 from the CAD routing), crimp tools, insulation tester, label printer, paint shop (outsourced) or a spray booth.

D Manufacturing

- **Shells:** order the STEP files from an MJF bureau (KAD3D / Formero AU [R6]). 1. **Before printing, add bosses:** Ø6 × 6 mm with a Ø4.2 hole for M3 heat-set inserts, every 100–150 mm along the edges, matching the frame standoffs. 2. Bead blast, or vapour smooth for a premium finish. 3. Fit-check every panel on the robot **before** painting. 4. Paint: adhesion promoter → 2K primer → satin warm white (25–30 GU) on the shells, matte graphite on the joint caps, soft clearcoat. 5. Mask the magnet and insert locations.
- **Light guides:** machined clear PC with diffuser film; amber LED strips (2.2 mm) behind.

E Assembly

1. **Harness:** - Build each zone harness on the nail-board. - Label both ends (zone-joint-function). - Test continuity 100 % and insulation at 500 V (> 100 MΩ). - Install through the hollow bores; P-clip every 80 mm. - Leave service loops at every joint (a 30 mm loop each side of each actuator).
2. **Cooling:** fit the torso and thigh ducts and filters. Run the blowers and check the airflow direction with a smoke pen.
3. **Bellows and gaiters:** waist bellows clamped to the pelvis and torso rims; ankle gaiters over the push-rods.
4. **Shells:** - Press heat-set inserts into the bosses. - Fit silicone isolators on the frame standoffs. - Mount the inner panels with screws, the service panels with quarter-turn studs + magnets. - Check the 1.0 ± 0.3 mm panel gaps with feeler gauges, and that no shell touches a moving part through the full range of motion (run the range slowly with the shells on).

5. **Final torque audit:** re-torque every actuator flange (KA-130 M5 8.5 N·m, KA-110/90/KH-25 M4 4.5 N·m) and apply paint-pen witness marks.
6. **Weigh:** full robot and each module. **If mass > 92.6 kg (the design mass), update `params.py` and re-run `calc/run_all.py` before commissioning.**

F Electrical

- Final insulation test: bus → chassis with the star bond lifted (> 10 MΩ at 500 V), then refit the bond.
- Light LEDs on the 12 V rail through the safety MCU (status protocol: docs/02 §7).
- Verify that every connector is IP2X (finger test) and latched.

G Inspection before commissioning

Check	Pass
Range of motion with shells on, all joints	no contact, gaps maintained
Mass and CoM (two-scale method in the zero pose)	mass ≤ 92.6 kg; CoM height 0.95–1.03 m
Thermal soak: stand idle 30 min hanging in the gantry, all systems on	all temperatures within limits
Water/dust: light spray on shells (IPX4-style hand spray, 5 min) with power **off**	no water inside the torso/actuators after opening
Documentation: build log complete, serials recorded, calibration files committed	—

P13 Software bring-up and commissioning hand-off

Gate G3–G9

This phase takes the assembled robot from "all joints move" to the walking and manipulation gates. Detailed test procedures and pass criteria are in docs/11. The architecture is in docs/07.

A Visual reference

- Software layers: `diagrams/software_stack.svg`
- Gates: `diagrams/test_gates.svg`
- Simulation model: `sim/kestrel_k1.xml` (MuJoCo, generated with the sized torque limits)

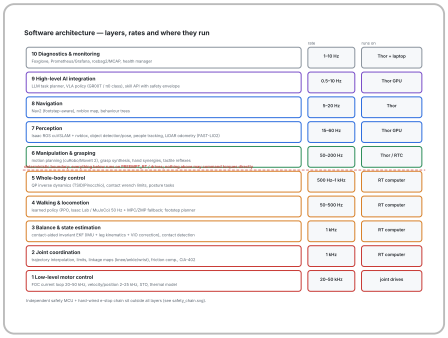


DIAGRAM Software layers



DIAGRAM Gates

B Software components

Component	Install
RT computer	Ubuntu 24.04 + PREEMPT_RT, IgH EtherLab 1.6 (igc), ros2_control (Jazzy), Pinocchio, ProxQP/OSQP, ONNX Runtime
AI computer	JetPack 7, ROS 2 Jazzy, Isaac ROS (cuVSLAM, nvblox), ZED SDK, Livox SDK2, Foxglove bridge
Safety MCU	monitor firmware (heartbeat, tilt, temperatures, e-stop, relay outputs)
Development PC	MuJoCo simulation, training package (see the training add-on), Foxglove

C Tools

Gantry + harness (P01), wireless e-stop, cage, foam floor, a second person, a laptop with Foxglove.

D Configuration files to produce (commit them)

1. `config/joints.yaml` : joint order, encoder offsets, directions, soft limits, torque limits (strain-wave classes at their repeated-peak values), linkage tables (`data/knee_fourbar.json` , ankle/wrist geometry).
2. `config/ethercat.xml` : slave order, PDO maps (CST mode: target torque, position, velocity; actual position ×2, torque/current, temperatures, status), DC settings.

3. `config/safety.yaml`: tilt limit 35°, temperature warn 130 °C / trip 150 °C, heartbeat 2 ms, comms timeout 50 ms.
4. `config/estimator.yaml`: IMU noise (CV7: 1.5 °/h bias instability, 30 µg/√Hz [ds R4]), kinematic noise, contact thresholds.

E Bring-up sequence (each step gated in docs/11)

1. **G3 hanging:** robot suspended, feet clear. Joint-by-joint at 10 % torque limits, then gravity compensation, then whole-body slow motions. Check every sensor stream and the estimator (base attitude vs the gantry inclinometer).
2. **G4 standing:**
3. Lower until the feet carry 20 % of the weight, then 50 %, then 100 % with the harness slack-taut.
4. Balance controller (WBC + IEKF).
5. Push tests ≤ 20 N·s.
6. Practise Cat 1 and Cat 0 stops caught by the harness.
7. **G5 stepping:** the ZMP walker (`calc/kestrel/gait.py` patterns, replayed through the WBC) for in-place steps, then 0.1–0.3 m/s, with the harness taking ≥ 10 % load.
8. **Learned policy:**
9. Train in simulation (training add-on).
10. Evaluate in MuJoCo against the ZMP walker on the regression suite.
11. Deploy with the harness, at 50 % action scale first.
12. **G6–G9:** free walking in the cage, stairs, endurance and manipulation per docs/11.

F Electrical / safety in this phase

- Every session starts with the e-stop functional check (press → drives disabled → reset).
- Torque limits are raised in steps (25 % → 50 % → 100 %) only after the previous level has run for 1 h without faults.

G Exit criteria

Gate G9 signed (docs/11): 1 h untethered walking at ≥ 0.8 m/s in the cage, 0.18 m stairs, 5 kg pick-and-place, all thermal margins logged, and a teardown inspection with no unexpected wear.

App. A Testing and commissioning gates

Nothing on KESTREL K1 is proven until it passes the tests below. Movement is introduced progressively: restrained → low speed → supported → free, with independent shutdown at every step.

1. Gate checklist (sign and date each)

Gate	Entry condition	Exit criteria (all required)
G0 Bench electrical	P02 complete	EtherCAT 1 kHz with distributed clocks, 1 h, 0 lost frames; STO < 10 ms from every e-stop device; safety-MCU trips verified; pack gate (P06 §G) passed
G1 Actuator	each actuator built	P03/P04/P05 §G table passed; measured Kt, η, thermal time constant and backlash fed back into `calc/` and the requirements re-checked
G2 Limb	limb assembled on its stand	P07/P09/P10 §G tables passed
G3 Hanging	robot assembled, suspended	every joint, sensor and estimator verified; Cat 0/1/2 stops tested; thermal soak OK
G4 Standing	G3	stand 10 min unassisted (harness slack-taut); recovery from 20 N·s pushes in 4 directions; sit/stand ×20
G5 Supported stepping	G4	100 steps in place, 50 m at 0.3 m/s with the harness carrying ≥ 10 % load; no falls caught by the harness in the last 20 m
G6 Free walking (cage)	G5	0.3 → 0.8 → 1.2 m/s; turns; slopes 5° → 15°; 1 h cumulative; harness slack throughout
G7 Stairs & terrain	G6	0.10 → 0.18 m steps up/down ×50 each; gravel, mats, cables on the floor
G8 Endurance	G7	2 h mixed duty with thermal margins logged; 10 h cumulative walking; teardown: gears, bearings, rod ends, bonds
G9 Manipulation & AI	G8	5 kg pick/place ×100; tool use; skill API rejects out-of-envelope requests; people exclusion zones enforced

2. Test environment and safe operating limits

- **Cage:** 4 × 3 m mesh panels, foam floor tiles, operator outside with the wireless e-stop on a lanyard. The gantry harness is attached from G3 to G5 and slack-attached in G6–G7.
- **Two-person rule** from G3. Nobody inside the cage while drives are enabled, and **no unrestricted high-torque operation around people** at any stage of this programme.
- **Limits until G6:** torque limits 50 % of the class peak, speed 50 %, workspace fences in software (base within the cage, hands below 1.8 m).
- **Abort criteria:** any unexpected noise or smell, temperature above warn, estimator divergence, EtherCAT errors > 0 per minute. Stop and investigate before continuing.

3. Structural validation

What	How	Pass
Hand calculations	`calc/kestrel/structure.py` (generated)	all SF ≥ 1.5 (structural case) / ≥ 2.0 (fatigue): **currently met**
FEA	pelvis bowl, yokes, knee hub, thigh plates, shank fittings, foot plate, ankle cross under N/M/X loads (CalculiX/FreeCAD FEM or Fusion)	von Mises < σy/1.5 (X); < endurance/2 (N); modes > 40 Hz (legs), > 25 Hz (arms)
Proof load	each leg on the stand: 1.5× the structural-case vertical load (≈ 3.3 kN) for 30 s, joints locked	no permanent set > 0.05 mm (dial indicators), no bond creep
Bonded joints	coupons: 1.5× axial + torsion	no failure
Fatigue	leg stand, 10^6 cycles of the walking load profile (≈ 140 h at 2 Hz) on one leg before G8	no cracks (dye penetrant on Ti and 7075 hot-spots)
Drop	the structural case (0.15 m) on the stand with a 93 kg sled	no yield; actuators may slip their flanges (inspect and re-torque)

4. Actuator validation (P03–P05 §G)

Item	Method	Feeds back into
Joint torque validation	dyno torque vs current across the speed range	`MOTORS` Kt, `ETA`
Motor thermal testing	thermal step at continuous torque to steady state; walking-duty replay from `data/scenario_series.npz`	`THERMAL_DERATE`; **the knee continuous margin (1.03) is the critical result**
Gearbox endurance	10^5 cycles (2×10^5 for KA-130) of the scenario torque profile; backlash and efficiency before and after	gear spec, maintenance interval
Efficiency map	speed × torque grid, motoring and regenerating	power budget (`calc/kestrel/power.py`)
Overload	locked-output impact at 1.5× momentary	flange slip setting

5. Electrical validation

Item	Method	Pass
Insulation	500 V bus → chassis (bond lifted); 1000 V motor windings → stack	> 10 MΩ / > 100 MΩ
Pack	P06 §G (capacity, balance, thermal at 1C, BMS trips, short-circuit on the disconnect)	as listed

Item	Method	Pass
Power quality	scope the bus during a W2 replay on the stand: ripple, regeneration peaks	bus stays 48–72 V; clamp engages above 70 V
eFuses	electronic-load trip per zone	±10 %
EMC pre-scan	near-field probe + LISN on the bus during walking; ESD ±8 kV air on exposed metal	no resets, no EtherCAT errors
E-stop response measurement	scope from the button contact to the phase current reaching zero, for each device; wireless out-of-range timeout	≤ 10 ms wired; wireless heartbeat timeout measured and logged (not published by the vendor [R2])
EtherCAT timing	log DC sync error and cycle jitter over 1 h with all 28 slaves	jitter < 20 µs, sync error < 1 µs

6. Sensor calibration and validation

See docs/06 §4 for procedures. Validation: walk a 20 m closed loop and compare the estimator against a motion-capture or total-station reference. Pass: drift < 1 % of distance without vision; < 0.2 % with VIO + LiDAR.

7. Control-system testing (progressive)

Step	Environment	What is tested
Software-in-the-loop	MuJoCo with `sim/kestrel_k1.xml` + the RT stack through ros2_control	stand, walk 0.8/1.2 m/s, stairs, pushes: regression suite before every hardware session
Single joint (G1)	dyno	torque, impedance, limits, faults
Single limb (G2)	leg/arm stand	tracking, gravity comp., load cases, linkage maps
Balance (G4)	standing, harness	WBC, IEKF, push recovery, controlled stops
Supported walking (G5)	harness ≥ 10 % load	ZMP walker first, then the learned policy at 50 % action scale
Progressive locomotion (G6–G7)	cage	speed, turns, slopes, stairs, terrain
Full-system endurance (G8)	cage	2 h duty, thermal margins, inspection

Sim-to-real checks: - compare measured joint torques (current × Kt) against the inverse-dynamics model for identical motions. Expected agreement ±15 %; investigate anything larger. - re-identify inertial parameters if the mass of any module deviates > 5 % from `params.py`.

8. Failure modes (FMEA summary)

Failure mode	Effect	Detection	Mitigation
Knee motor overheats in sustained walking	derate → slow, then Cat 1 stop	winding NTC + thermal model	ducted cooling; liquid-jacket fallback; gait with lower knee torque
Strain-wave ratcheting on impact	arm position loss	dual encoder mismatch	firmware torque limits; momentary rating 1.9× repeated
Push-rod or rod-end failure	knee/ankle loss of drive → fall	encoder mismatch (actuator vs joint)	buckling SF 5.5; inspection interval; harness during testing
Flange slip	joint offset	dual encoder mismatch	slip = overload fuse; re-zero + re-torque procedure
EtherCAT cable break	loss of slaves	master WKC error	ring redundancy; Cat 1 stop if the ring is broken in two places
Cell internal short / thermal runaway	fire	BMS (dT/dt, voltage), smoke	per-cell fuse links, mica barriers, disconnect, supervised charging, battery-area storage
RT computer crash	loss of control	safety-MCU heartbeat	Cat 1 → STO; harness
Estimator divergence	fall	covariance monitor, IMU vs kinematics residual	Cat 1 stop
Foot slip	fall	load-cell CoP jump, foot IMU	conservative friction cone (μ = 0.5 assumed vs 1.0 sole); speed limits
Hand overload	finger damage	current	breakaway detents, back-drivable screws
AI requests unsafe motion	collision	skill-API envelope check	requests are advisory; envelope enforced in the RT layer

9. Uncertainties and assumptions register

#	Assumption	Value used	How verified
1	Mass growth	+10 % (design 92.6 kg)	weigh modules; re-run `run_all.py`
2	KM-130 Km, Kt	model ±5 % vs two datasheets	dyno (P05)
3	Thermal derate with ducted air	1.00 (KA-130); 0.85 others	thermal step tests
4	Planetary efficiency	0.92	dyno
5	Strain-wave efficiency	0.75	dyno

#	Assumption	Value used	How verified
6	Voltage-limited speed factor	k_L = 0.75 (calibrated to TQ rated speed)	torque-speed sweep at 52 V
7	Gait model	flat-footed ZMP with dynamics filter (conservative knee); learned gait differs	measured torques vs model
8	Minimum loaded bus	52 V (3.25 V/cell)	BMS derating curve; low-SOC walking test
9	Prices	84 % estimate/quote	quotes before ordering
10	Unverified components	DC-DC bricks, fuses, STO signal levels, CubeMars RI50, wireless e-stop response time	datasheets/quotes before design freeze

10. Reporting

Each gate produces a short report in `logs/gates/Gx.md`: test list, data plots, deviations, decisions, and the updated calc outputs. Keep failed attempts too; they are the most valuable data.



DIAGRAM Test gates

App. B Manufacturing package

1. What is in the package and its status

Asset	Location	Status
Full robot assembly (240 parts, named)	`viewer/kestrel_k1.glb`, generated by `cad/build_robot.py`	**envelope CAD**: correct sizes, positions, interfaces and clearances for packaging; not every hole is modelled
Manufacturable part models (STEP + STL)	`cad/out/step/`, `cad/out/stl/`	shells, frames, tubes, rods, brackets: **printable/machinable as envelopes**; add fastener holes from §4 before ordering
Detailed load-path parts	`cad/out/step/detail__*.step` (ankle cross, foot plate, thigh plate, knee hub)	**fully detailed**: bores, bolt patterns, pockets, lightening
KA-130K / KH-25 actuator parts (full revolve)	`cad/out/step/ka130k__*.step`, `kh25__*.step`	turned-part profiles: housings, rotor hub, carriers, flange, ring-gear blank
Drawing sheets (projected view + title block: material, process, tolerance, finish)	`cad/out/drawings/*.svg` (19 sheets)	visual reference; dimensions come from the STEP
Gear, lamination and winding specifications	§5–§6 below, docs/13	specification level

The honest status: this is a **complete concept-to-preliminary design package**. Before cutting metal, each envelope part goes through a detailing pass that adds holes, threads, fillets and datum features. The four detailed parts above show the standard to reach. Then FEA (docs/11 §3).

2. General tolerances and standards

Item	Standard / value
General machining tolerances	ISO 2768-mK (medium linear, K geometric) unless noted
Bearing seats (rotating outer ring)	housing **H6/N6** per bearing maker; shaft **k5** (inner ring rotating) or **j5**
Crossed-roller seats	per THK/IKO catalogue for the chosen preload class; housing H6, shaft h5/k5; seat runout ≤ 0.01 mm
Dowel holes	Ø4 / Ø6 **H7**, dowels m6 (17-4PH or hardened steel)
Gear ring spigots	H7/js6

Item	Standard / value
Bonded joints (CF tube in 7075 fitting)	radial bond gap 0.10–0.15 mm, 25 mm engagement, 3M DP490 / Loctite EA 9466, grit-blast + primer
Threads	ISO metric coarse, 6H/6g; helicoil (stainless) every tapped hole in aluminium that is reassembled (actuator flanges)
Surface finish	bearing seats Ra 0.8; gear bores Ra 0.8; general Ra 3.2; anodised faces Ra 1.6 before anodise
Edge break	0.2–0.5 mm on all edges unless noted
Fastener preload	VDI 2230; M4 12.9 ≈ 6.9 kN, M5 12.9 ≈ 9.9 kN at 70 % proof (μ_thread 0.12); torque with a calibrated wrench, Loctite 243 on actuator flanges

3. Processes by part family

Family	Material	Process	Finish	Notes
Actuator housings (26)	7075-T6 bar	CNC lathe + 3-axis mill (fins turned)	Type III hard anodise black 25–40 μm; **mask** bearing seats, bonding pads and the stator bore	anodise adds 50 % inward: machine the critical bores after anodise, or mask them
Pelvis bowl, yokes, brackets, spine, ribs	7075-T651 plate/billet	3- or 5-axis CNC	hard anodise black	stress relieve not needed (T651)
Thigh/foot plates	7075-T651 8–12 mm plate	CNC (or waterjet + CNC finish)	hard anodise	flatness 0.1
CF tubes	roll-wrapped CFRP Ø40 × 2 (thigh/shank), Ø50 × 2 (upper arm)	cut with a diamond blade, ends sanded	clear coat	Enhanced Composites (AU) or Rock West
Ti parts (push-rods, crosses, pins)	Ti-6Al-4V bar/tube (AMS 4928)	CNC turning/milling, sharp tools, flood coolant	bead blast	PCBWay CNC lists Ti Gr5 [R6]
Gears	18CrNiMo7-6 (or 17-4PH for lightly loaded stage 1)	hob/shape (external), **wire-EDM or gear-shaping** (internal rings) → carburise 0.6–0.8 mm case, 58–62 HRC → grind teeth (ISO 1328 grade 6)	—	gear job shop to drawing

Family	Material	Process	Finish	Notes
Shells (≈ 42)	PA12 HP MJF	MJF print (KAD3D, Formero AU [R6])	vapour smooth or bead blast → 2K primer, satin top coat	wall 2.0–2.4 mm; M3 heat-set inserts in bosses
Visor	PC 3 mm	thermoform over a printed buck, or machine	smoke tint + AR hard coat on apertures	
Bellows/gaiters	TPU 95A	FDM print or urethane cast	—	
PCBs	FR-4 4-layer 2 oz (power), 6-layer (hand controller)	JLCPCB / PCBWay assembly	conformal coat (acrylic)	IPC-A-610 class 2
Battery enclosure	6061-T6	CNC	clear anodise	mica liners
Fingertip caps	Dragon Skin 20	cast in printed two-part moulds	—	dovetail onto a PA12-CF core

4. Interface standards (use on every envelope part when detailing)

Interface	Standard
KA-130 output flange	PCD 105, 12× M5 12.9 (helicoil), 2× Ø6 H7 dowels at 180°, spigot Ø90 h6
KA-130 base flange	PCD 125, 12× M5, 2× Ø6 dowels
KA-110P / KA-90P output flange	PCD 85 / 75, 10× / 8× M4 12.9, 2× Ø4 H7 dowels, spigot Ø60 h6
KH-25 / 20 / 14 output flange	PCD 70 / 58 / 42, 8× M4 / 8× M3 / 6× M3, 2× Ø4 dowels
Hollow cable bores	Ø20 (KA-130), Ø15 (KA-110/90), Ø12 (KH)
Shell bosses	M3 heat-set insert, Ø6 boss, 3 mm silicone isolator
Joint zero pins	Ø4 mm H7 hole through link + actuator housing at the zero position

5. Custom gears (KA family)

Gear	Module	Teeth	Face width	Profile	Material / HT	Grade
KA-130 stage-1 sun	0.8	20	14	20° PA, x = +0.3	17-4PH H900 (or 18CrNiMo7-6 case-hardened)	ISO 6
KA-130 stage-1 planets (3)	0.8	15	12	x = −0.3	same	ISO 6
KA-130 stage-1 ring	0.8	50	14	internal	42CrMo4 Q&T 30 HRC (EDM-cut)	ISO 7

Gear	Module	Teeth	Face width	Profile	Material / HT	Grade
KA-130 stage-2 sun	1.25	18	18	x = +0.3, tip relief	18CrNiMo7-6 carburised 60 HRC	ISO 6
KA-130 stage-2 planets (4)	1.25	18	16	x = −0.3, crowned	same	ISO 6
KA-130 stage-2 ring	1.25	54	18	internal	18CrNiMo7-6 carburised + ground, or 42CrMo4 nitrided	ISO 6
KA-110P (32:1) / KA-90P (23:1)	0.8 / 0.7	per detail design (compound stepped planets)			as above	ISO 6

Planet pins: 17-4PH H900 Ø6 (stage 1) and Ø10 (stage 2) with needle rollers. Carrier: 17-4PH H1025. Lubrication: high-viscosity synthetic grease (NLGI 1–2, e.g. Klüber Isoflex Topas class) with O-ring sealed housings. Assembly backlash target: ≤ 10 arcmin at the output, measured by the dual encoders.

6. Motor manufacturing specification (summary; full procedure in docs/13)

Item	KM-130x45 / KM-130x40	KM-50x20
Laminations	0.20 mm NO20 (or 0.35 M270-35A), laser-cut or wire-EDM stack, bonded (Backlack or glued), burr ≤ 0.02 mm, insulation C-5	0.20 mm
Stack	45 / 40 mm, 36 teeth, OD 130 h7, ID (tooth tips) 97.4	20 mm, 12 teeth
Slot insulation	Nomex 410 0.13 mm, Class H	same
Winding	double-layer concentrated, 5 t × 2 paths (x45) / 3 t × 1 (x40), multi-strand grade 2 enamel Class H 200 °C	18 t, Ø0.69 equivalent
Impregnation	vacuum dip in Class H varnish, bake	same
Magnets	42 × N42SH arcs, 3.0 mm thick, 45 mm long, magnetised radially in alternating polarity; dimensional tolerance ±0.05	14 arcs, 2.0 mm
Rotor	1018 back-iron ring (machined), magnets bonded with high-temperature epoxy + 0.3 mm CF/Kevlar sleeve, balanced G2.5	same
Tests	R (4-wire), L, back-EMF constant, hi-pot 1,000 V AC 1 min winding-to-stack, locked-rotor torque/current, cogging	same

7. Electronics manufacturing

- **Drive carrier** (26), **PDU**, **disconnect board**, **safety MCU board**, **hand controller** (2), **foot board** (2): 4–6 layer FR-4, 2 oz copper on power layers, ENIG finish, JLCPCB/PCBWay

assembly. Every board gets a test pad set and a scripted bring-up test (docs/10 P2).

- **Harness:** build on a printed nail-board from the CAD routing; label every wire; 100 % continuity and insulation-resistance test (500 V, > 100 MΩ) before installation.

8. Battery pack manufacture

See docs/10 phase P6 (step-by-step) and docs/05 §3. Build in a cleared area with a sand bucket and a Class D (lithium) or large CO_2/water mist extinguisher, never alone.

9. Inspection plan

Part class	Inspection
Bearing seats, gear bores	bore gauge / CMM, 100 %
Gears	gear report from the supplier (profile, lead, runout) + incoming double-flank check
Flatness-critical plates	surface plate + dial indicator
Ti parts	dye-penetrant crack inspection after machining
Bonded tube joints	proof load 1.5× structural-case axial + torsion on a test coupon from each batch
Shells	fit check on the assembly before painting
Cells	100 % mass, OCV and 1 kHz IR; reject outliers (R3)

App. C Optional: scratch-built motors and actuators

This track replaces bought motors and gearboxes with parts you make: wound copper stators, magnet rotors, cut gears, machined housings and your own drives. It is **optional**. The baseline (docs/04) already builds the knee and hip-pitch motors (KM-130) and the KM-50 from scratch, because no commercial part meets those requirements. Everything here extends that to the rest of the robot.

Generated designs: generated/km_scratch_designs.md (one motor per actuator class, with coil-by-coil winding tables) and generated/km_motor_designs.md.

1. What can and cannot be made: feasibility matrix

Part	Make at home	Make with a job shop	Buy only	Why / closest buildable alternative
Stator **laminations**	× (needs laser/EDM + insulation-coated electrical steel)	✓ laser-cut or wire-EDM stacks from NO20 / M270-35A sheet		Order prototype stacks to your DXF; bonding is done at home
Stator **winding** (copper coils)	✓	✓		Hand needle-winding on the teeth with a jig is practical for 36 slots
Magnets (sintered NdFeB)	×	×	✓	Sintering, grinding and magnetising NdFeB needs an industrial process (hydrogen decrepitation, jet milling, aligned pressing, vacuum sintering at ~1,080 °C, pulse magnetisers). **Buy custom-shaped arcs** (N42SH, NiCuNi coated) from magnet suppliers. Bonded-NdFeB (printable) has about half the remanence and would cost ~40 % of the torque.
Rotor back-iron, hubs, shafts	✓ (lathe)	✓		1018 steel ring, 7075 hub
Rotor **sleeve**	✓			wet-wound carbon or Kevlar tow on a mandrel
Bearings (crossed-roller, deep-groove, needle)	×	×	✓	Precision-ground races and rollers; buy

Part	Make at home	Make with a job shop	Buy only	Why / closest buildable alternative
Planetary gears	△ (external gears on a CNC with a hob/4th axis, prototypes)	✓ hob/shape, wire-EDM rings, carburise, grind		Gear shop to drawing (docs/09 §5). Printed gears are for fit checks only
Cycloidal reducer (alternative to strain-wave)	△	✓ wire-EDM discs, hardened pins		**The recommended DIY replacement for strain-wave gears** (§4.2)
Strain-wave flexspline	×	× (specialised)	✓	Thin-walled (≈ 1 % of diameter) alloy-steel cup with fine involute teeth that flexes 2× per input turn. Fatigue life depends on special steel, heat treatment, shot peening and tooth form. Not reproducible reliably. Use **cycloidal** instead, or buy
Housings, carriers, flanges	✓ (mill/lathe)	✓		7075; hard anodise (Type III) outsourced; Type II you can do at home
Encoders	✓ magnetic (MT6835 chip on your own PCB + diametric magnet)		✓ optical/inductive absolute rings (AksIM-2)	Making the absolute ring/scale is not practical; the IC is a buy-and-solder part
Motor drives	△	✓ PCB assembly		Feasible from the TI TIDA-010979 architecture (§7). Months of firmware and safety work
MOSFETs, gate drivers, MCUs			✓	semiconductors
Cells	×	×	✓	Pack assembly is yours (P06)
Hand micro-motors (10 mm)	×	×	✓	Winding 10 mm coreless motors (self-supporting skew windings) needs dedicated machines; buy maxon/Faulhaber
Ball screws (micro)	×	×	✓	ground thread + recirculating balls

Part	Make at home	Make with a job shop	Buy only	Why / closest buildable alternative
Ti push-rods, crosses	✓ (lathe/mill, care with Ti)	✓		

✓ feasible · △ possible with significant tooling/skill · × not realistically possible

2. The scratch-built motor family

Motor	Replaces	Use
KM-130x45 / KM-130x40	(baseline custom)	knee / hip pitch
KM-85x26	TQ ILM85x26	KA-110P (hip roll, waist pitch)
KM-85x17	CubeMars RI85-PH	KA-90P (hip yaw, waist yaw, ankles) and KH-20
KM-85x12	TQ ILM85x13	KH-25 (shoulders)
KM-50x20	(baseline custom)	KH-14

The three KM-85 designs share **one lamination die** (Ø85, 36 slots) and **one magnet arc** (42 × 7.29° × 2.5 mm). Only the stack length and the turns change, which makes them cheap to make in quantity. The model reproduces the commercial motors they replace within ±6 % on Km, so the actuator sizing in docs/04 stays valid if you substitute them. **Re-run `calc/run_all.py` with the measured values once built.**

3. Making a motor, step by step (KM-130x45 shown; the others are identical in method)

3.1 Verify the design in FEMM first (free, 2-D magnetostatic)

1. Draw one **symmetry section**. For 36s/42p the pattern repeats every 12 slots / 14 poles, so model 1/3 of the motor (12 teeth, 14 magnets) with anti-periodic/periodic boundaries.
2. Materials: NO20 (or M270-35A) BH curve, N42SH at 80 °C (Br ≈ 1.22 T, μr 1.05), copper, air.
3. Check, at rated current (34.8 A rms → peak in phase A, half in B and C), stepping the rotor 0.5° over one pole pair: * airgap flux density fundamental ≈ 1.2 T at open circuit (the model says 1.20 T); * tooth and yoke peak flux ≤ 1.8 T at rated current, and the knee of the curve at ≈ 3× rated (saturation → peak torque); * torque constant within ±10 % of 0.310 N·m/A rms; * cogging torque peak-to-peak < 1 % of rated (36s/42p has LCM 252, so cogging is inherently low).
4. Mechanical: the calculated yoke (2.1 mm) is magnetically sufficient but fragile. **Use a 3.0 mm yoke** (stator OD unchanged, slot depth −0.9 mm, about −6 % slot area, accepted) and confirm the flux densities again.

3.2 Laminations

1. Export the lamination DXF from the FEMM geometry. Features: - 36 teeth; tooth tips with a 1.2 mm deep shoe and a 2.0 mm slot opening; - 3 mm yoke; OD 130.00; - 2 key notches on

the OD for housing alignment; - 2 small weld or bond notches.

2. Order a **prototype stack** from a lamination service (laser-cut + stacked + bonded, or wire-EDM a pre-bonded block): 0.20 mm NO20 with a C-5 insulation coating. Stack height 45.0 mm ≈ 225 laminations. Burr ≤ 0.02 mm. Ask for a stacking factor ≥ 0.95.
3. If you receive loose laminations, bond them yourself:
4. Clean them.
5. Stack on a hardened mandrel with key bars (keeps the slots aligned).
6. Press 1 MPa in a fixture and wick Loctite 290 (or use self-bonding Backlack laminations heated per the supplier).
7. Cure in an oven.
8. Check that the stack is square (≤ 0.05 mm) and that a slot gauge passes through every slot.
9. Deburr the slot edges with a fibre brush. Check insulation between laminations (> 1 kΩ across the stack with a megger at 50 V). That limits eddy losses.

3.3 Slot insulation

1. Cut **Nomex 410, 0.13 mm** liners to the slot perimeter plus 2 mm (a printed template helps). Crease and insert one per slot, folded over at the tooth tips.
2. Fit end-lamination insulators (printed PA12-CF or Nomex) on both stack faces so the wire never touches lamination edges.

3.4 Winding (the copper coil)

Use the coil table in generated/km_scratch_designs.md:

```
36 slots / 42 poles, coil k around tooth k ('+' clockwise seen from the lead end):
T01:+A T02:+B T03:-B T04:-C T05:+C T06:+A T07:-A T08:-B T09:+B T10:+C T11:-C T12:-A   (repeats ×3)
```

- **Conductor:** KM-130x45 needs 5 turns × 2 parallel paths of ≈ 1.78 mm equivalent copper per turn. Wind **7 strands of Ø0.67 mm grade-2 enamelled copper, Class H (200 °C)** in hand, so the bundle stays flexible and AC losses stay low. KM-130x40: 3 turns of ≈ 2.30 mm equivalent (12 × Ø0.67).
- **Method (needle winding by hand):** 1. Mount the stator on a rotary indexing jig (printed, 10° detents). 2. For each tooth: anchor the bundle at the lead end, then wind the required turns around the tooth in the table direction, laying turns flat and tight against the liner. 3. Use a smooth nylon wedge to dress each turn into the slot. Each slot holds two half-coils (double layer), so keep each coil within its half of the slot. 4. After each coil, tie it off with Kevlar lacing at both end faces. Label the start/finish leads (A1s, A1f…).
- **Connection:**
- Each phase has 12 coils.
- KM-130x45 (2 paths): two groups of 6 coils in series per phase (alternating teeth of that phase in the sequence), the groups paralleled.
- Observe the signs: a "−" coil is connected finish-to-finish with its neighbour (equivalent to winding it the other way).

- Join the three phase ends in a **star** with a crimped and soldered star point, insulated with Class H sleeving.
- **Leads:** 3 phase leads in high-temperature silicone (12 AWG for KM-130), plus 2 NTCs (EPCOS B57861S) laid against the end turns at the hot spot (near the lead exit).
- **Check before varnishing:**
- phase resistances equal within ±2 % (KM-130x45 model: 32.6 mΩ line-to-line at 20 °C);
- insulation > 100 MΩ at 500 V to the stack;
- a surge/comparison test if available, otherwise the phase inductances must be equal within ±3 % on an LCR meter.

3.5 Impregnation

1. Pre-heat the stator to 80 °C.
2. Vacuum-dip (or trickle-impregnate) in **Class H polyester-imide or epoxy varnish**.
3. Drain, then bake to the varnish schedule (typically 2 h at 150 °C).

This fills the voids (thermal conduction from copper to iron, mechanical locking, moisture protection). Re-test insulation afterwards.

3.6 Rotor

1. **Back-iron ring:** - 1018 steel, OD 89.6 (the magnets add 2 × 3.0 to give a rotor OD of 95.6 mm), radial thickness 3.6 mm; - turned and ground to concentricity ≤ 0.01 mm; - press or bond it onto the 7075 hub, which has its two bearing seats.
2. **Magnets:** - Order 42 arcs: N42SH, 3.0 mm thick, 7.29° arc at Ø95.6 OD, 45 mm long (or 3 × 15 mm segmented axially, which cuts eddy losses and is easier to source). - NiCuNi coated, **radially magnetised**. Order 10 % spares. - Check every magnet's polarity with a gaussmeter and mark the N face.
3. **Bonding fixture:** - A printed or aluminium (non-magnetic) indexing ring with 42 pockets holds the magnets in **alternating polarity** (N out, S out, …) at exact pitch. - Bond with a high-temperature structural epoxy (Loctite 9492 / 3M 2216 class). - Load one magnet at a time. Each new magnet is attracted strongly to the steel ring, so slide it in from the side with a non-magnetic pusher and **never put fingers between a magnet and steel**.
4. **Sleeve:** - Wet-wind carbon tow (or Kevlar) 0.3 mm thick over the magnets with epoxy, under light tension. - Cure and grind to the final OD (airgap 0.60 mm to the stator bore). - This retains the magnets at speed and in a crash.
5. **Balance:** static on knife edges, then dynamic on the lathe at 1,000 rpm using a phone-accelerometer app, removing material from the hub to G2.5.

3.7 Assembly (rotor into stator): magnet safety

The radial pull of a 42-pole rotor entering a stator is several hundred newtons and acts suddenly. 1. Mount the stator (potted in its housing, P05 §E) vertically on the press table. 2. Fit a **Ø20 mm hardened guide rod** through the housing's bearing seat and the rotor hub bore. A printed PTFE sleeve on the rotor keeps it concentric. 3. Lower the rotor with the press quill (never by hand) until the bearings seat. 4. Remove the sleeve, fit the second bearing and the wave spring, then rotate by hand: smooth, with faint cogging only.

3.8 Bearings and encoder

Rotor bearings (68xx thin-section) j5/H6, wave spring 30–60 N preload. Motor encoder: - **AksIM-2 ring** on the hub, readhead on the rear cap; or - **MT6835** on the drive PCB facing a diametric magnet at the hub end (shaft-end mounting). A hollow hub needs an off-axis magnetic ring, so use the AksIM-2 ring for KA-130.

3.9 Cooling options

- Baseline: potted stator in a finned 7075 housing with ducted air (derate 1.00).
- **Liquid jacket (fallback if the knee thermal test fails):**
- a helical 6 × 3 mm channel machined in a stainless or aluminium sleeve, shrink-fitted over the stator OD and sealed with O-rings;
- 0.3–0.5 L/min water–glycol from a 12 V micro pump (TCS / Bürkert class);
- a radiator in the thigh duct.
- Expected continuous gain ≈ 1.6× (copper-to-coolant thermal resistance ≈ 0.25 K/W vs ≈ 0.6 K/W air-cooled; validate on the dyno). Adds ≈ 0.35 kg per leg.

3.10 Motor acceptance tests (before building it into an actuator)

Test	Equipment	Pass (KM-130x45 model)
Phase resistance (4-wire) at 20 °C	milliohm meter	32.6 mΩ L-L ±10 %, phases within ±2 %
Inductance	LCR meter, 1 kHz	phases within ±3 %; record Ld/Lq by rotor position
Hi-pot, winding → stack	1,000 V AC (or 1,500 V DC) 60 s	no breakdown, leakage < 1 mA
Insulation resistance	500 V DC	> 100 MΩ
Back-EMF constant	spin with a second motor at 500 rpm, scope L-L voltage	Ke L-L peak 0.253 V·s/rad ±7 %, sinusoidal (THD < 5 %)
Torque constant	locked rotor on a lever + load cell, 5–35 A	Kt 0.310 N·m/A rms ±8 %; linear to ≈ 80 A, then saturating
Cogging	rotate slowly with a torque arm	< 0.1 N·m peak-to-peak
Thermal step	10.8 N·m equivalent current (34.8 A) in the potted housing with duct air	winding ≤ 120 °C at steady state

If Kt or Km is more than 10 % off, update `calc/kestrel/actuators.py` (or `motor_design.py`'s calibration) and re-run `calc/run_all.py` before building more.

4. Making the gearboxes

4.1 Planetary (KA family)

- **Recommended:** a gear shop machines to the docs/09 §5 table (hob/shape, carburise, grind).
- **Prototype path at home:** 1. Print the gears in PA12 or tough resin to check meshing, backlash and assembly inside the rotor. 2. Then CNC-hob the external gears in 17-4PH on a 4th axis (single-tooth fly cutter or a module hob on a mill arbor) for low-load tests. 3. Internal rings: wire-EDM from 42CrMo4 pre-hardened plate (job shop). 4. Heat treatment (carburise + temper, or nitride) is always outsourced.
- Inspection: over-pin measurement (two pins in opposite tooth spaces) to verify tooth thickness; meshing backlash 0.03–0.06 mm at m1.25.

4.2 Cycloidal reducer: the DIY replacement for strain-wave gears (KH family)

Recommended scratch-build arm actuator: **KH-C** (cycloidal), replacing the bought strain-wave sets.

Item	KH-C25 (shoulder)	KH-C20 (shoulder yaw, elbow)
Ratio	30:1 (31 ring pins, 30-lobe disc)	30:1
Ring-pin circle	Ø84 mm, pins Ø5 mm hardened dowels (62 HRC) in a 42CrMo4 ring	Ø68, pins Ø4
Eccentricity	0.9 mm (K1 = E·N/R = 0.66; min epitrochoid curvature radius 4.9 mm vs pin radius 2.5 mm)	0.8 mm (K1 0.73; 3.7 mm vs 2.0 mm)
Discs	2 discs at 180° (balance), 8 mm thick each, 42CrMo4 Q&T + nitrided, **wire-EDM profile** (generated from the standard cycloid equations with a 0.02 mm profile modification for clearance)	6 mm
Output	8 pins Ø8 through disc holes into the output flange (crossed-roller)	6 pins
Expected	backlash 1–3 arcmin with precision EDM; efficiency 0.82–0.88; momentary capacity > 3× rated	
Effect on sizing	30:1 instead of 50:1: speed +67 %, torque × 0.6 × (0.85/0.75). KH-25 → ≈ 119 N·m peak (req. 67), 11.8 rad/s; KH-20 → ≈ 128 N·m (req. 55), 12 rad/s	

The disc profile for EDM is generated by `cad/cycloid_profile.py` (outputs `cad/out/cycloid_KH-C25.dxf`, `cycloid_KH-C20.dxf` and STEP discs) from the standard equations, with pin-circle radius R, pin radius Rr, eccentricity E and N ring pins:

```
psi(t) = atan2( sin((1−N)t),  R/(E·N) − cos((1−N)t) )
x(t)   =  R·cos t − Rr·cos(t + psi) − E·cos(N·t)
y(t)   = −R·sin t + Rr·sin(t + psi) + E·sin(N·t)
```

The script checks E·N < R and that the pin radius is below 0.8× the minimum radius of curvature of the pin-centre epitrochoid (no undercut). It also removes 0.02 mm for running clearance. Verify the meshing in CAD before cutting. Cycloidal reducers tolerate shock better than strain-wave gears and need no special steel. They are heavier, about +0.15 kg per actuator.

4.3 Strain-wave (if you insist)

Not recommended (§1). If you experiment, treat it as research. The flexspline fatigue life cannot be validated without long endurance testing and metallurgy.

5. Housings and structure

Everything in docs/09 §3 is CNC work you can do yourself with a capable mill and lathe (7075, Ti with care). Hard anodise (Type III) is outsourced. Type II sulphuric anodise is possible at home with a small tank, but it is softer.

6. Bearings, magnets, cells, semiconductors

Buy them (see §1). Use authorised distributors for cells (counterfeits are common, R3) and for magnets (request grade certificates with Br/Hcj at temperature).

7. Building the joint drives yourself (cost-down option)

- **Architecture (TI TIDA-010979 pattern [R2]):**
- AM2612 or F28P650DK MCU (the F28P65 has an on-chip EtherCAT slave controller and SIL2 hardware);
- 3 × DRV7167 100 V GaN half-bridges (60 A rms), or DRV8353RS + 6 × ISC022N10NM6 MOSFETs;
- isolated delta-sigma current sensing (AMC0106M05) or INA241 + 1 mΩ shunts;
- 2 × BiSS-C encoder interfaces; EtherCAT PHYs;
- a Ø70 mm annular board with a 15 mm bore.
- **Firmware:** FOC (Park/Clarke, SVPWM 20–40 kHz), current/velocity/position loops, field weakening, CiA-402 state machine over EtherCAT (ESI file), encoder calibration, thermal model, fault handling.
- **STO:** two independent hardware channels that remove gate-driver supply/enable, so they cannot be overridden by firmware. Test the response time.
- **Effort:** 6–12 months for a robust, tested drive. Recommendation (R2): buy modules first and design your drive as a later cost-down for the low-power joints.

8. Cost and time comparison (per actuator class, rough)

Item	Buy route	Scratch route	Notes
Ø85 frameless motor	A$450–1,300 (CubeMars/TQ)	A$180–350 (laminations, magnets, wire) + ≈ 15 h	big saving on the TQ classes

Item	Buy route	Scratch route	Notes
KM-130 motor	n/a	A$1,050–1,100 + ≈ 25 h	baseline already
Strain-wave set	A$450–640 (Leaderdrive)	cycloidal A$250–400 (EDM + pins) + ≈ 10 h	cycloidal is more robust
Drive	A$945–1,600	A$150–250 parts + months of development	after the robot works
Encoder (motor side)	A$456 (AksIM-2)	A$18 (MT6835)	baseline already uses MT6835 on 19 joints

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Generated from `docs/10-construction-tutorial/`, `docs/09`, `docs/11` and `docs/13` of the KESTREL K1 design package by `make_handbook.py`. The engineering numbers come from `calc/run_all.py`; when a design value changes, re-run it and regenerate this page.